

Crying Wolf, Carefully: What It Takes to Evaluate Paraphrase Attacks on Sampling-Based Hallucination Detectors

ScriptSampler

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Abstract

Sampling-based hallucination detectors such as semantic entropy are deployed on a claim of phrasing-invariant uncertainty. We tried to break it with meaning-preserving paraphrases and found the binding constraint falls on the operator, not the attacker. The *discrete* estimator we study weights meaning clusters by sample counts, so it is the entropy of a partition of N : at the standard $N=10$ it takes 39 values, with an atom at the maximum. No threshold above the maximum fires, so while the atom carries mass the lowest false-alarm rate a firing threshold can have *is* the chance a clean correct answer yields N mutually distinct meanings. Under NLI clustering that is 10.5% [9.0, 12.2] on 1424 clean correct answers. At that standard $N=10$ an operator cannot have a 5% false-alarm budget, however well the score ranks; both qualifiers are load-bearing. At $N=40$ none of the fair pool’s 200 correct answers reaches $\ln 40$ (0/200, at most 1.9%). The atom is empty, and the obstruction with it. What 200 answers cannot do is locate the floor: their cheapest alarm costs 2.0%, but that is the population’s floor only if it can never yield 39 distinct meanings in 40, a question 200 answers cannot settle, so we quote no interval. The threshold honouring a 5% budget achieves 5.0% [2.7, 9.0] on the same 200 answers. We report that as a measured operating point, not a demonstration that one at or below 5% exists. Measuring an attack here needs stronger controls than the literature’s random-perturbation baseline: re-scoring each selected paraphrase on an independent sample retains only 44% [23%, 64%] of the apparent effect. Our protocol makes it answerable: score-independent target selection, an answer-invariance success criterion, a null priced against the attacker’s search budget, and an independent LLM-judge equivalence oracle. The confirmatory run is complete and does not reject; we claim no attack effect here.

1 Introduction

Large language models are increasingly paired with *hallucination detectors* that estimate whether a generated answer can be trusted. A prominent family is sampling-based. Semantic entropy (SE) [6, 12] samples several answers, clusters them by meaning with a natural-language-inference (NLI) equivalence check, and reports an entropy over the resulting clusters. The literature carries three such estimators, and Section 3 separates them; we study the *discrete* variant, which weights clusters by sample counts. Semantic reformulation entropy (SRE) [28] extends the idea with input reformulations. These detectors are deployed on an implicit promise: that the uncertainty they report is a property of the *question and answer*, invariant to how the question happens to be phrased.

That invariance has been asserted but not adversarially tested *on the detector itself*. Prior red-teaming of LLMs targets the model’s answer, either by eliciting hallucinations [14, 15] or by evading text-origin classifiers through paraphrase [3, 5]. Work on uncertainty manipulation has used fine-tuning or backdoors rather than a meaning-preserving input change [31], and the closest detector-side attack degrades a hidden-state probe model-side [18]. Uncertainty *is* attackable from the input side: Obadinma and Zhu [21] show that semantics-preserving perturbations degrade an LLM’s *verbal* confidence. We therefore scope the gap narrowly. What remains untested is the *sampling-based* family, whose score is not a self-report but a statistic of an answer distribution partitioned by a

learned equivalence relation. Its attack surface is accordingly that sampling-and-clustering pipeline rather than a model’s self-assessment. The question we ask is therefore whether an attacker who only *rephrases the question*, under a hard semantic-equivalence constraint, can move a sampling-based detector’s score in a chosen direction while the model’s answer stays put, and its correctness with it.

Answering this question rigorously turns out to require controls that, to our knowledge, prior evaluations of this detector family do not jointly apply. Once those controls are in place, the picture is more subtle than a clean “we broke it,” and the result we lead with is not about the attacker at all. Building the controls forced us to ask what a false-alarm rate can even be on a detector whose score takes finitely many values, and the answer to that question turned out to be the finding. We contribute the controls as a protocol and report what they reveal. Three of them are load-bearing. (i) Targets are selected *score-independently*, since otherwise the clean detector is evaluated on a pool its own scores defined, a circularity that inflates clean AUROC to 1.0. (ii) An attack counts only if the model’s *hallucination status* is unchanged under the paraphrase, checked with the same oracle that labelled the data. (iii) Effect sizes are reported *net of a benign-paraphrase noise floor*, since the detector score is a finite-sample estimate and an optimiser selecting the extremum over many candidates otherwise mistakes sampling variance for success.

We do now report on whether the optimised attack moves the detector beyond what benign rephrasing achieves, and the answer is that it does not. The pre-committed exceedance test has a null that prices the attacker’s search budget in. Run to completion over the 80 correct-answer targets of the false-alarm stratum, that test fails to reject in every clustering arm, at $p=0.939$ (shared NLI), 1.000 (exact match) and 1.000 (LLM judge) on the 77 of those targets that have a benign arm, each a median over 101 tie-break draws. Those p -values price in the 181 scored evaluations the optimiser performs per target, and we report 181 as an upper bound on the budget rather than as the budget: the attack’s maximum ranges only over candidates its equivalence gate admitted, the gate is applied lazily so most scored candidates were never submitted to it, and nothing the run recorded identifies how many of them would have passed. The shared-NLI arm’s p moves with that unidentified quantity. The pre-registered adjudicator arm does not, and it is the arm we rest the non-rejection on: its p holds at 1.000 across the whole admissible range of the budget (Section 4). This is a non-rejection by a design with power 0.77 against a two-fold effect, so it bounds what we could have seen rather than measuring a zero, and we claim no attack effect from it.

The arm the optimiser actually searched is in any case the detector’s own NLI clusterer. That is precisely the measure that cannot separate a change in the model’s answer distribution from an inconsistency of the NLI model that the detector’s clustering and the attack’s equivalence gate *share*. The independent oracles one would reach for to break that tie are sentence embedders, and they turn out to be near-chance on exactly the word-preserving, meaning-shifted case an attack produces. On current evidence, then, the attribution is *unresolved but bounded*, and single-clusterer paraphrase-attack claims on these detectors are not adequately supported. We formulate the attack in *both* directions as the case study that exercises the protocol. The *hide* direction is designed to suppress a wrong answer’s score, and the *false-alarm* direction to inflate a correct answer’s. Only the false-alarm direction is evaluated under the null control in this work.

The three controls above were built to measure that attack honestly, and building them is what surfaced the result we lead with. A null that has to stay valid when an attacked score and a benign one tie at the detector’s maximum forces one to characterise that maximum, and characterising it shows that clean correct answers reach it often enough, at the sample budget the detector is deployed with, to put a floor under the false-alarm rate of every threshold rule that fires at all. That floor is the measurement-validity result below. We arrived at it by trying to measure an attack, not by setting out to audit the estimator, and we say so here because the order matters for reading what follows: the protocol is not scaffolding around the attack that we then reused, it is where the finding came from.

Contributions. (1) **A measurement-validity result about *discrete* semantic entropy**, which the attack investigation uncovered and which stands whether or not any attack succeeds. The scope is deliberate and is the black-box deployment path: the discrete variant is the one Farquhar et al. [6] name and run for all of their GPT-4 results, the one Kossen et al. [11] train their probes to predict, and the only one computable under the black-box access we assume (Section 3). The two halves

of the result have different reach, and we keep them apart. That a plug-in entropy over at most N observed clusters cannot exceed $\log N$ is elementary, and we claim no novelty for it; our contribution is the measurement of *how tightly that bound binds in deployment*. That half is shared with the likelihood-weighted Eq. (5), which normalises to a categorical distribution over the same clusters and so inherits the same ceiling. It is not shared with Kuhn’s unnormalised Eq. (4), which has no such bound, so the ceiling is not a property of “semantic entropy” unqualified. The lattice half below is discrete-specific and does not transfer.

At the standard $N=10$ the estimator is confined to a lattice of 39 attainable values with an atom at the maximum $\ln 10$, and the consequence we measure is a constraint on the *operator* rather than on the attacker: a detector that flags when the score reaches a threshold can only change its false-alarm rate at a value some clean correct answer actually took, so the rates it can be run at are a short finite list. Scored clean on the fair pool’s 200 correct answers, the achievable clean false-positive rates below one in four are 0%, 9.5% [6.2, 14.4] and 21.5% [16.4, 27.7], with nothing in between; an operator who specifies a 5% false-alarm budget cannot have one at that sample budget, because the only threshold that honours it never fires. Running the same 200 correct answers out to $N=40$ empties the ceiling atom and buys the missing operating point, so a 5% budget is honoured at an achieved 5.0% [2.7, 9.0]. We therefore scope the claim to $N=10$, and to the detector’s own bidirectional DeBERTa-large-MNLI clustering, wherever we make it (Section 5).

That floor is an identity, not a measurement we happened to make: every threshold above $\ln N$ flags nothing, so the smallest false-positive rate a firing threshold can have *is* the probability that a clean correct answer yields N mutually distinct meanings. The identity binds only while the cap carries mass. At $N=10$ it does, on the direct cache. At $N=20$ it does as well, but that figure is a *replay*. The $N=40$ run’s samples subset down to 20, which by exchangeability is an unbiased estimator of a direct $N=20$ pass but is not one: no direct measurement of the grid at $N=20$ exists, the only answers this project has ever scored at that budget being the 15 saturated targets of the headroom pilot below, which were selected on their attacked $N=10$ score. At $N=40$ the identity does not bind: no clean correct answer reaches $\ln 40$, the at-cap mass is 0.0% [0.0, 1.9], the floor reverts to an ordinary order statistic, and the cheapest threshold that fires at all costs 2.0% rather than nothing. We attach no interval to that 2.0%; Section 5 says why.

We put that probability at 9.5% on the fair pool’s correct stratum and, on the 1424-answer superset that stratum is a subset of, at 10.5% [9.0, 12.2], whose lower bound closes off a 5% budget at 95% confidence; a 10% budget sits on the boundary, and the two populations land on opposite sides of it, so we claim only that the floor is about one correct answer in ten. That floor is a fact about the equivalence relation deciding when two answers mean the same thing, and not about the sampler alone. Holding the same clean samples fixed and varying only that relation moves the at-cap rate by more than an order of magnitude, across a range that straddles the 5% line, and the exact-match and NLI intervals do not overlap (Section 4). We accordingly state the floor under the clusterer the detector actually ships and claim its oracle-dependence, rather than a floor that holds under whatever relation a deployment happens to cluster with; the looser oracle we validate puts too little mass at the cap to be distinguished from 5% in its own right, which is a reason to scope the claim and not a reason to weaken it.

The claim is about *deterministic* rules, which is what a deployment can be audited against: a randomised rule that flags a ceiling item with some probability does reach the chords between these points, at the price of paying an indivisible block’s average, and below the floor the reachable frontier is a single chord out of the origin. Nothing here rests on a count of distinct values realised: that quantity is monotone in the number of targets scored (the fair pool’s correct stratum yields 28 at $n=200$ and the full labelled pool 35 at $n=2000$, and 22 sits at the 37th percentile of what a random 80 drawn from that correct stratum produces), so it measures the sample, not the estimator, and the total number of operating points inherits the same defect. What does not move with the sample is the lattice size, and at the low end the floor. The general argument has been made and named for black-box LLM classification [27]: a score’s distinct-value count caps what an operator can do with it, however well it ranks. We instantiate that argument for a sampling-based generative detector, where the granularity limit is imposed by the sample budget itself.

The originating work sees the same crowding from the other side, reporting that clusters are few

and grow little with N , and reads it as a computational saving [6, 12]. Read instead as a measurement property, it means false-alarm effects are *right-censored* on just over half of attacked targets, so native-unit effect sizes understate them and any null must stay valid under ties at the cap. This sits consistently with reports that the plug-in estimator *underestimates* true semantic entropy at practical N [17, 23], and we stop at consistency. Two mechanisms would produce that understatement and nothing we measure separates them: truncation at $\log N$, since a statistic that cannot exceed the cap must understate whenever the quantity it estimates is larger, and the finite-sample negative bias of a plug-in entropy over a semantic alphabet the sample under-covers, which is the mechanism those two papers propose and correct for. They are not mutually exclusive and both are active at deployed N , so we claim agreement with their finding rather than an explanation of it. (2) **A measured account of how much of a paraphrase-attack effect is an artefact of its own search.** Re-scoring each *selected* paraphrase on an independent sample, we find only 44% of the apparent effect survives (95% CI [23%, 64%]). The two statements this supports have very different strengths and we keep them apart: *that* the selection-time figure is inflated is established, since the shrinkage is -0.341 nats with a 95% CI of $[-0.469, -0.212]$ that excludes zero; *how much* is not, since the retention interval runs from under a quarter to two-thirds. Roughly half of what an uncorrected report would claim is selection on sampling noise. That attack success scales with the search budget is known [8], as is the case that success-rate comparisons need explicit measurement validity [4]; what we add is the measurement for a *stochastic detector score*, where the randomness being resampled lives inside the estimator rather than in the victim’s decoding. (3) **A protocol that makes the question answerable:** score-independent target selection, an answer-invariance success criterion, and an exceedance test that prices the attacker’s search budget into the null analytically, with tie handling that remains valid when the score saturates. Without score-independent selection the clean detector looks perfect, reaching AUROC 1.0 by construction, versus 0.704 [0.653, 0.753] on the fair pool. Each control is individually precedented; we are not aware of them applied jointly to a sampling-based hallucination detector. That hedge carries a stated coverage limit rather than an implied one. Section 6 enumerates the venues our search could not reach, including indices we verified return nothing to control queries. Absence of hits in those indices is therefore uninformative rather than weakly reassuring, and this claim is bounded accordingly. (4) **The independent equivalence oracle the attribution requires.** Because the detector’s NLI model both clusters answers and certifies paraphrase equivalence, a single-clusterer evaluation cannot separate a real change in the model’s answers from that model’s own inconsistency. Cheap oracles do not resolve it. Sentence embedders are near-chance on exactly the adversarial case, so we build and validate an LLM-judge that clears that discrimination (0.93 vs. 0.51). The confirmatory evaluation under it has run, and under that adjudicator the exceedance test does not reject; the pre-registered $n \geq 80$ was met by the attacked stratum but not by the test, which ran on the 77 of those targets with a benign arm (Section 4). (5) **The attack itself as the case study that exercises the protocol,** together with a reformulation-averaging defense and an analysis of what it does and does not close. The attack is bidirectional and NLI-constrained, and it foregrounds the induced-false-positive direction. We claim neither the direction nor the search primitive as new: inducing a detector false positive while the underlying prediction stays correct is an established family [10, 26], and constrained paraphrase search is a standard tool [14, 15]. What is not inherited is the *evaluation*, because of three properties of this victim class. First, and most structurally, semantic entropy’s score *is* an equivalence judgement: it counts how many distinct meanings a sample set contains. The decision that produces it is the same kind of semantic-equivalence decision the attack’s gate must make, and in our case it is made by literally the same model. The effect is therefore unattributable without an oracle independent of the detector. That confound cannot arise for a constraint given by an ℓ_p ball, which is model-free and definitional, nor for a classifier, whose score is not an equivalence relation at all. Second, the score is a finite-sample estimate, so a maximum over candidates inflates: we measure 44% retention on re-scoring. This does not arise against the deterministic classifiers targeted in the work above, though it would against any stochastic scorer. Third, the estimator is *discrete*, and coarsest exactly where the attack must land. Boundedness alone would not set it apart, since classifier confidence is bounded too. At $N=10$ the estimator’s attainable lattice has 39 points, and its top two lie 0.139 nats apart, which is less than the success criterion’s own δ , so 21 of our 80 correct targets cannot register a success at any usable δ , whatever the paraphrase.

Raising N relaxes this, and substantially: $N=20$ affords 455 points, seven of them in the top tenth of the range. What a lifted cap returns in *usable headroom* is a different quantity and the only one of the two we measured, at $+0.304$ nats against a nominal $+0.693$ on a pilot of 15 saturated targets, selected on their *attacked* $N=10$ score. We report that measurement without a mechanism for it (Section 5). Effect sizes are therefore right-censored and any null must stay valid under ties at the cap. The discreteness half of this is the score-granularity gap named by Sun et al. [27]; what we add is its instantiation for a sampling-based detector, where the lattice is set by the sample budget and is at its coarsest at the top of the scale that both a false alarm and an operator’s threshold must reach. What we claim is what this victim class forces on the measurement, not the move itself. No effect size entered the body until the definitive run landed under the budget-matched null control, which it did on 2026-08-29 (Section 4); the small-sample machinery pass is treated as pipeline-validation only.

2 Related Work

Sampling-based hallucination detection. Semantic entropy estimates a model’s uncertainty by sampling several answers, clustering them into meaning-equivalence classes, and taking an entropy over those classes [6, 12]. How the classes are *weighted* is not a detail, and the originating papers do not settle on one answer: Kuhn et al. [12] average cluster surprisals under length-normalised sequence likelihoods, Farquhar et al. [6] normalise those likelihoods into a categorical distribution and take its plug-in entropy (their Eq. (5), which they designate as their main method), and the same paper introduces a third, *discrete* variant that weights each cluster by the number of generations in it. The three do not share their measurement properties, so we name ours: we study the discrete variant, for reasons and with consequences set out in Section 3. Its central selling point is *linguistic invariance*: because clustering is on meaning rather than surface form, the score is meant to be stable across rephrasings. Subsequent work improves the estimator. Semantic Reformulation Entropy (SRE) adds input-side reformulations and energy-based hybrid clustering for robustness [28], and Semantic Entropy Probes (SEPs) approximate the entropy from a single hidden state, removing the sampling cost [11]. A parallel line questions the *clustering* step itself: Kernel Language Entropy replaces hard NLI equivalence-class clustering with a positive-semidefinite semantic kernel scored by a von Neumann entropy, on the ground that a hard partition is too *coarse* to carry the pairwise semantic dependencies between answers and so returns uncertainty estimates less fine-grained than those dependencies support [20]. We are careful not to enlist that as agreement with us, because it is a different objection. Theirs is about what a hard partition can *represent*; ours is that the relation drawing the partition need not be stable under a rephrasing of the question, and separately that the entropy of a hard partition of N samples can take only finitely many values. The three concerns land on the same step from different sides, and a graded relation answers the third by construction: a von Neumann entropy over a continuous-valued kernel is not confined to the entropies of integer partitions of N . We treat these as *detectors to be attacked*: our contribution is to test the invariance claim adversarially, a question none of these papers ask of themselves. Whether the attack transfers to such richer-clustering variants, or is specific to hard-NLI-equivalence clustering, is an open question our null-controlled protocol is designed to pose.

Hallucination-elicitation attacks on models. The closest methodological neighbours attack the *generating model*, not a detector. SECA frames realistic hallucination elicitation as constrained optimization over the prompt under semantic-equivalence and coherence constraints, solved with a zeroth-order method [14]; its successor REALISTA extends this to free-form reasoning models via continuous latent edit directions [15]. We reuse SECA’s optimization loop as tooling, but our objective is the *detector’s* uncertainty score rather than the model’s answer correctness, and we add a direction (false alarm) absent from both. Crucially, neither evaluates a hallucination detector, so neither subsumes our result despite the shared machinery.

Paraphrase attacks on text-origin detectors. A mature line paraphrases text to evade AI-versus-human classifiers [3, 5]. Wang et al. [30] give the cleanest formalisation of the primitive, casting

detector-evasive paraphrasing as a constrained Markov decision process in which semantic preservation is an explicit constraint rather than a scalarised reward term. These share the high-level move “paraphrase to fool a detector” but differ on every axis that matters here: the victim is a text-origin classifier, not a sampling-based uncertainty estimator; the attacks are evasion-only (one-directional); and the equivalence guarantee is a soft similarity score rather than the hard bidirectional-NLI gate we require. We adopt a distinct name to avoid confusion with Cheng et al. [3].

Manipulating uncertainty and probes. The closest input-side neighbour is Obadinma and Zhu [21], who show that semantics-preserving perturbations degrade an LLM’s *verbal* confidence. That work carries the same group’s calibration-attack line [22] from vision into language. The victim differs in the way that matters here. Verbal confidence is the model’s prompted self-report, whereas semantic entropy is computed from a sampled answer set clustered by meaning, so the attack surface is the sampling-and-clustering pipeline rather than a self-assessment. We also impose a bidirectional-NLI equivalence gate they do not, and our success criterion *requires* the model’s answer status to survive the paraphrase, whereas their attacks frequently change the answer. Also on the “manipulate uncertainty” axis, Zeng et al. [31] shift answer confidence via a fine-tuned backdoor trigger on multiple-choice self-evaluation, which is a different threat model altogether: weight access rather than input paraphrase.

We should be explicit that the *direction* we foreground is not itself new. Moving a confidence signal in an attacker-chosen direction while holding the underlying prediction fixed is an established family with a name: Obadinma et al. [22] give what they describe as the first dedicated study of *calibration attacks*, which “trap victim models to be heavily miscalibrated without altering their predicted labels”. The family is still active, and Martínez-Martínez et al. [16] propose an underconfidence attack that lowers confidence while preserving the prediction, paired with an adversarial-training defence. Forcing a detector into a false *positive* has a direct text-modality precedent: Rusert [26] perturb text so that an attack-detection model flags an input while the classifier’s prediction stays correct. Concurrently, Khanmohammadi et al. [10] manipulate deployed confidence channels bidirectionally under an answer-preservation constraint, including an inflation direction, and report the same qualitative collapse of a gated system toward its ungated floor. We therefore do not claim the false-alarm direction as a discovery. What we claim is its instantiation against a *sampling-based* detector, which differs on two axes that carry consequences rather than labels. First, the manipulated quantity is an entropy over meaning-clustered samples rather than a softmax or a learned detector’s logit: a stochastic finite-sample estimate, bounded above by $\log N$, produced by the same NLI model that certifies the attacker’s constraint. Second, the perturbation is a question rewrite rather than a pixel budget or a hidden-state intervention, and a deployed adversary possesses neither of those channels. Semantic invariance therefore cannot be granted by an ℓ_p ball and must instead be certified. We accordingly treat the equivalence constraint as an instrument requiring its own validation rather than as a definition: we constrain, re-check that answer invariance actually held, measure how often the gate nonetheless admitted a non-equivalent rewrite and report that as a disclosed lower bound, and validate the oracle making the judgement on an adversarial discrimination that disqualified the obvious cheap candidate (0.51). An attack whose constraint is an ℓ_p ball has no analogue of these steps, because there the constraint is definitional rather than inferred. Most relevant of all, CORVUS red-teams single-pass hallucination detectors and degrades SEP [18], which establishes that SE-*family* detectors can be attacked. We do not claim that as our novelty. Our distinction is substantive rather than a label. CORVUS perturbs a frozen hidden-state probe *model-side* (LoRA fine-tuning, fixed inputs, suppression-only, no semantic-equivalence constraint, Alpaca/FAVA). We perturb the *input* under a hard equivalence constraint, in both directions, on TriviaQA. Our target is the sampling-and-NLI-clustering pipeline itself, which is a different attack surface. A weaker neighbour prompts LLMs to fool generic activation probes [2], but not in the hallucination-detection setting. Each ingredient of our attack is by now standard. Constrained search over meaning-preserving paraphrases is an established primitive, and Kaneko [9] run essentially our optimiser loop, searching paraphrases that raise an LLM reviewer’s score under semantic-equivalence and naturalness constraints. Directional manipulation of a confidence signal under prediction-preservation is an established family. What does not carry over is the *evaluation*. Our victim’s score

is a stochastic, ceiling-bounded estimate produced by the same model that certifies the attacker’s constraint, and such a victim invalidates the standard protocol in three specific ways: selection on estimator noise, right-censoring at the cap, and unattributability under a shared clusterer. None of the three arises for a classifier victim. That is where we locate our contribution, and we evaluate the false-alarm direction here while building the hide direction as its mirror. The bidirectional-NLI equivalence gate is our design choice for certifying meaning-preservation rather than a novelty axis in its own right. Our own analysis in fact treats the detector’s reliance on that same NLI model as the central confound.

What the estimator can and cannot express. A separate line studies semantic entropy as a *statistic* rather than as a detector. McCabe et al. [17] show that the discrete plug-in estimator underestimates the true semantic entropy at practical sample sizes, because the semantic alphabet is under-covered, and correct it upward; Pan et al. [23] pursue the same correction with a coverage-adjusted estimator fusing Good–Turing coverage with a spectral term over an entailment-weighted response graph. We characterise the same estimator’s opposite tail. It cannot exceed $\log N$, and in deployment it piles up there. That two groups now correct the estimator upward is itself evidence for our reading rather than against it: a remedy at the level of the *estimator* concedes that the deployed plug-in statistic is the limiting factor, which is precisely our claim. The two accounts are complementary rather than competing, and we are careful not to promote that into an explanation. Two mechanisms can make a plug-in estimate fall short of the semantic entropy it targets: truncation at $\log N$, which binds whenever the true quantity exceeds the cap, and the finite-sample negative bias of a plug-in entropy over an alphabet the sample under-covers, which is the mechanism these papers correct for. They are not mutually exclusive, both are active at deployed N , and nothing we measure apports the understatement between them, so we claim agreement with their finding and not a diagnosis of it. Separately, Sun et al. [27] name the *score granularity gap* for black-box LLM classification: a confidence score may rank well and still take so few distinct values that an operator has only a handful of usable thresholds. *Discrete* semantic entropy inherits this pathology at the budget it is deployed at, with 39 attainable values at the standard $N=10$ and two of them in the top tenth of the range. The twist is that the granularity limit is set by the sampling budget rather than by a verbalisation format, and is therefore relaxable by spending in a way a verbalisation format is not: $N=20$ affords 455 values, seven of them in that top tenth. We accordingly assert the claim at $N=10$ and nowhere else, and what carries to larger N is only that the lattice stays sparsest at the top of the scale, which is where a false alarm and an operator’s threshold both have to land. Our contribution here is the instantiation and its consequence for attack evaluation, not the argument itself. The likelihood-weighted estimator does not inherit it: real-valued cluster weights are not confined to the entropies of integer partitions of N , so the lattice dissolves even though the $\log N$ ceiling remains (Section 3). The granularity claim is thus the sharper and the narrower of our two. It is narrower in the estimator it holds for and in the sample budget it is asserted at, and we keep the two claims apart wherever we make them. The originating papers report that clusters are few and grow slowly with N , and treat that as a computational saving [6, 12]. Taken together with those reports, the picture is of a well-known observation whose measurement consequences have not been drawn out.

Achievable operating points and randomised rules. That a decision rule built on a discrete statistic can be run at only finitely many operating points, and that the gaps between them are closed by randomising rather than by a better score, is classical, and we claim none of it. It is why randomised tests exist. The Neyman–Pearson formulation fixes a size and maximises power at it [19]; when the statistic is discrete a nonrandomised test attains only finitely many sizes, so the standard treatment states the fundamental lemma for test functions taking values in $[0, 1]$, randomising the decision at the critical value, precisely so that a nominated level stays attainable exactly [13]. The classification-side statement is the ROC convex hull. Provost and Fawcett [24] identify the classifiers on the hull of a family’s ROC points as exactly those potentially optimal under some combination of class distribution and misclassification costs, and realise a point interior to a hull segment by flipping a weighted coin, per instance, between the two classifiers at its ends; Fawcett [7] states the same interpolation in textbook form. Distribution-free prediction meets the arithmetic from a

third direction: conformal prediction [29] calibrated on n scores has its coverage pinned between $1 - \alpha$ and $1 - \alpha + 1/(n+1)$, a band whose width is set by the calibration sample rather than by the operator’s target, and holding its upper end under ties takes an explicit randomised tie-break that practice usually skips [1]. Our claim is therefore not that a finite operating grid is a new phenomenon. It is that this grid is imposed by the *sampling budget*, so an operator cannot refine it by labelling more evaluation data as they could refine a calibration set; that it is coarsest at the top of the scale, which is exactly where a false alarm and a firing threshold both have to land; and that the atom at the very top puts a floor under the false-alarm rate of every deterministic rule, which is a constraint the classical results permit but do not predict, since none of them fixes where the atoms fall. Randomising does reach the chords here as it does there, at the price of paying an indivisible block’s average and of a rule whose response to a given input cannot be audited against a fixed threshold, and we say so wherever we make the claim.

Evaluating the evaluation. Because our framing is protocol-led, two methodological strands matter beyond the attack neighbours. First, the individual failure modes our controls target are classical: selection on the evaluated system’s own scores, selection of an extremum of a noisy estimate, and success criteria that drift from the construct. So are the individual remedies. Comparing an adversarial perturbation against a random one of equivalent magnitude is established practice in vision robustness evaluation [25], and plain-paraphrase baselines are routine in AI-text-detector evasion [3]. We claim neither as new. That an uncontrolled baseline can manufacture an entire effect is not hypothetical: Zheng and Kurban [33] report an apparent gain of up to 31.8 points from a perturbation-based selection rule that disappears once a format- and budget-matched control is introduced, the control tracking or exceeding the method on every benchmark. What we are not aware of, for a paraphrase attack on a sampling-based hallucination detector, is the *joint* application of score-independent selection, answer-invariance checking, and a noise floor that is *budget-matched* to the attacker’s own search. That floor is the quantity the attack’s maximum is actually compared against. Our contribution is assembling these into a pre-registered protocol with a decision rule, and we state the gap as “to our knowledge” rather than as priority. Second, using an LLM as an equivalence oracle inherits the LLM-as-judge literature’s lesson that judges track humans well yet carry biases and must be validated per task [32]. Our judge contribution is accordingly not the concept but the *adversarial-case* validation: the oracle is accepted only on the hard-negative discrimination where the cheap embedding oracle demonstrably fails (0.93 vs. 0.51), with its pre-registered acceptance gate, the failed positive-recognition leg, and the consequent false-alarm-only re-scoping disclosed rather than assumed away (Methods).

3 Methods

Detectors, and which semantic-entropy estimator we study. Given a question q , the target model M samples N answers $\{y_1, \dots, y_N\}$ at temperature T . Bidirectional natural-language inference partitions them into semantic-equivalence clusters $\mathcal{C}(q)$: y_i and y_j share a cluster iff each entails the other under the NLI model. What is done with that partition differs across the originating literature in a way that decides which of our claims travel, so we distinguish three estimators before naming the one we study.

(a) *The Monte-Carlo estimator of Kuhn et al. [12]* sets $p(c \mid q) = \sum_{y \in c} p(y \mid q)$ over *length-normalised* sequence likelihoods and takes their Eq. (4),

$$\text{SE}(q) \approx -|\mathcal{C}|^{-1} \sum_{i=1}^{|\mathcal{C}|} \log p(C_i \mid q),$$

an average of cluster *surprisals* with no renormalisation across clusters. This is not a categorical entropy over the observed clusters and it is *not* confined to $[0, \log N]$: it diverges upward as any cluster likelihood tends to zero, and it is not bounded below by zero either, since length normalisation can lift cluster masses above one. Renormalising would not rescue the upper bound but invert it, since

Jensen’s inequality places a uniform average of $-\log P(C_i | q)$ at or above $\log |C|$, with equality only for a uniform cluster distribution. None of this is a defect of Eq. (4) on its own terms: it targets the entropy of the model’s full semantic distribution, a quantity that genuinely may exceed $\log N$, and being unbounded is what lets it do so. The point is only that a bound argued for one estimator may not be quoted for another.

(b) Equation (5) of Farquhar et al. [6] supplies the missing normalisation,

$$P(C_i|x) = P(c_i|x) / \sum_c P(c|x).$$

The stated grounds are that “[w]ithout this normalization step cluster ‘probabilities’ could exceed one because of length normalization, resulting in degeneracies,” and the estimator then takes the plug-in entropy of the resulting categorical distribution. They are explicit about its status: “Equation (5) is the estimator giving our main method that we refer to as semantic entropy throughout the text.”

(c) *Discrete semantic entropy*, the estimator we study, discards the likelihoods and weights each cluster by the number of generations in it, giving the Shannon entropy of the cluster-size distribution:

$$H_{SE}(q) = - \sum_{c \in \mathcal{C}(q)} p(c) \log p(c), \quad p(c) = \frac{|c|}{N}.$$

We study the discrete variant throughout, for three reasons. First, it is not our simplification but the victim’s own: for “scenarios in which the sequence probabilities are not available,” Farquhar et al. [6] “propose a variant of semantic entropy which [they] call ‘discrete’ semantic entropy,” which “approximates $P(C_i|x)$ directly from the number of generations in each cluster”. It also carries their flagship closed-model result, being the estimator “which we use for all GPT-4 results in this paper and which performs similarly well.” Second, it is the quantity the probe line is trained to predict: Kossen et al. [11] “use an additional approximation, employing the *discrete* variant of SE that yields good performance without access to token probabilities, making it compatible with black-box models,” so a result about the discrete estimator is a result about what SEPs regress on. Third, it is the only one of the three computable under the black-box threat model we declare next, which grants query access but not token probabilities: an adversary or auditor with that access has no likelihood weights to use.

This is a narrowing rather than a retreat, but it is a real narrowing and we mark the line here rather than leave it to be inferred. Claims resting only on the *bound* transfer to Farquhar’s Eq. (5): it is a normalised categorical entropy over the same at most N observed clusters, so $H \in [0, \log |C|] \subseteq [0, \log N]$ by the same argument, and with it travel the existence of a ceiling, the possibility of right-censored false-alarm effect sizes, and the requirement that a null remain valid under ties. They do *not* transfer to Kuhn’s Eq. (4), which is unbounded above; the ceiling therefore cannot be asserted of “semantic entropy” unqualified, and writing “discrete” once would not be enough. Claims resting on *discreteness* are ours alone. They are the 39-point lattice at $N=10$, the 0.139 nats between its top two points, the unwinnable share of the false-alarm denominator, the exact-saturation counts, the hard-tie handling in the exceedance test, the finite grid of achievable false-alarm rates, and the non-zero floor beneath it. All of them presuppose integer cluster sizes and lapse under real-valued weights, and the last two are the most consequential. Eq. (5) attains its own maximum only when the normalised cluster likelihoods are exactly equal, an event of measure zero, so it has no ceiling *atom* and hence no floor on the false-positive rates it can be operated at; the ceiling is shared, the atom that makes the ceiling an operating constraint is not.

We state the consequence for the main method plainly, where the scoping is introduced, rather than leave a reader to derive it from the argument, because it is the least favourable reading of our own result and it is now measured rather than argued. We re-scored all 2000 cached sample sets of our replication pass under Eq. (5), using the generating model’s own length-normalised sequence likelihoods and holding the clustering fixed, so that the weighting rule is the only thing that varies. Those sample sets are at natural prevalence over the 1424-answer clean-correct superset and its complement, not the fair pool. Under Eq. (5), 0.0% [0.0, 0.2] of the 2000 cached sample sets fall at the cap, against 14.8% [13.3, 16.4] under the discrete estimator. **The estimator Farquhar et al. [6] designate as their main method does not exhibit the atom at all**, and the false-alarm floor the

atom induces is therefore a property of sample-proportion weighting rather than of semantic entropy. Three things bound how far that reaches and none of them undoes it, so we give them here and not only in mitigation. The reweighting preserves the ordering. It gives Spearman $\rho=0.99$ against the discrete estimator, so every AUROC in this paper stands under Eq. (5) *as we compute it*, and only the saturation arithmetic is estimator-specific. The qualifier is not a formality: length normalisation is the modelling choice that decides this, and on raw sequence likelihoods the cluster softmax is nearly one-hot and rank agreement with the discrete estimator collapses to $\rho=0.38$ (Section 6). Near-cap *crowding* partially survives: 20.0% [18.3, 21.8] of the same questions still fall in the top tenth of the range under Eq. (5), against 27.8% [25.9, 29.8] under the discrete estimator, so the range is genuinely crowded near its top under both and only the discrete estimator turns that crowding into a floor. And one ceiling-derived claim is invariant to the weighting altogether. Because every variant scores the *same* clustering and differs only in how the clusters are weighted, and because $H \leq \log K$ for any normalised weighting over K observed clusters, a score in the top tenth of the range $[0, \log N]$ requires $K \geq \lceil N^{0.9} \rceil$ distinct meanings. At $N=10$ that is $K \geq 8$, since $\log 8 = 2.079$ exceeds $0.9 \log 10 = 2.072$ while $\log 7 = 1.946$ does not. That is a necessary condition on the clustering, which the estimators share, rather than on the weighting, which is what they disagree about; on the fair pool it is met by 58 of 200 correct answers (29.0% [23.2, 35.6]) against 115 of 200 hallucinating ones (57.5% [50.6, 64.1]), so for most of the correct stratum the top of the scale is unreachable under any weighting rather than merely unreached under ours. The bound follows from the $\log N$ ceiling, so it too is silent about Kuhn’s Eq. (4). Two disclosed gaps ride along with the re-scoring and we do not paper over them. First, the cache stores decoded answer strings rather than the token ids the sampler emitted, so teacher forcing scores each answer under its *canonical* tokenisation. In the cache, 17 of 20,000 samples retokenise past the generation budget, which proves the two tokenisations differ for at least those. Second, the sampling manifest pins neither the quantisation type nor the model revision. The full estimator comparison is in Section 6.

Semantic Reformulation Entropy (SRE) [28] first generates R paraphrases of q , draws K samples from each, pools all $R \times K$ answers, and computes a single entropy over the joint clustering. A higher score is read as higher uncertainty; a detector flags an answer as a likely hallucination when its score exceeds a threshold.

Threat model. The attacker is black-box: it may query M , the detector, and an auxiliary LLM paraphraser, but has no gradients. It may replace the question q with any paraphrase q' subject to a hard equivalence constraint, $\text{NLI}(q, q')$ and $\text{NLI}(q', q)$ both entailment, so q' is constrained to ask the same thing. This gate *approximates* answer-invariance but does not guarantee it formally. Bidirectional entailment between two questions is an NLI model’s judgement, not a proof. That is exactly why the success criterion below spends a separate condition re-checking the answer empirically. The model’s answer is not otherwise constrained; the attack manipulates only the *detector’s score*.

Attack objectives. We define two directions. The *false-alarm* attack targets a question the model answers correctly and seeks to *inflate* the detector score so a correct answer is flagged: $\max_{q'} H(q')$ subject to the equivalence constraint. The *hide* attack targets a question the model answers incorrectly and seeks to *suppress* the score so a hallucination is missed: $\min_{q'} H(q')$, i.e. $\max_{q'} -H(q')$. Writing both as maximization of a signed objective lets a single optimizer serve both, with the sign encoding the direction. The same objective is instantiated against either detector ($H \in \{H_{\text{SE}}, H_{\text{SRE}}\}$), which is what makes the cross-detector transfer measurement well-defined.

Optimizer. We adapt the constraint-respecting zeroth-order search of SECA [14]. The search maintains a beam of the top- P paraphrase candidates ($P=3$); each round proposes $B=3$ further paraphrases per beam member, scores them with the objective, keeps those that improve on the running best, and admits a candidate only if it passes the bidirectional-NLI gate against the *original* q rather than the parent. The paraphrase proposer is the victim model itself (Llama-3.1-8B-Instruct) decoding *greedily*; candidate diversity comes from a seeded randomisation of the paraphrase instruction (verb,

style, and template drawn per call), so a campaign’s candidate set is reproducible. Reusing the victim as paraphraser adds no access assumption beyond the black-box query access already assumed, and any capable paraphraser would serve. Anchoring every candidate to the original means equivalence error cannot *accumulate* across rounds; it does not eliminate it. Because the optimizer searches many candidates for the one that maximizes the objective while passing the gate, it will preferentially surface single-hop NLI false positives. The residual fidelity of that one gate hop is therefore not assumed away but measured directly, in the answer-flip subcategory of the evaluation below. The search terminates on a score threshold or an iteration budget of 20 rounds, which is $1 + 20 \times P \times B \approx 181$ objective evaluations per target. That count is the number of candidates *scored*, and it is an upper bound on the attacker’s search budget rather than the budget itself: only a candidate that also clears the feasibility gate can become the reported maximum, and the gate is applied lazily, so the number of candidates that maximum actually ranges over is only partly identified. Section 4 bounds it to [41, 181], reports the test across that range, and rests the verdict on the arm that is invariant over it. The benign noise floor of $m=50$ random NLI-passing paraphrases is matched against that budget analytically, by the exceedance test of that section, rather than by setting m equal to it. We swap SECA’s objective (confidence in a wrong multiple-choice token) for the detector entropy above, and its LLM-judge feasibility check for the NLI gate.

Target selection. Both attacks require a pool of questions with a known clean outcome. The hide attack needs wrong answers and the false-alarm attack needs correct ones. We label correctness by greedy decoding under an alias-aware span-match oracle (canonical normalization, crediting any accepted answer form) and draw a *score-independent*, stratified-random sample within each correctness stratum. Selection reads only the correctness label and a fixed seed, never any detector score; consequently the SE and SRE campaigns operate on the *identical* question set by construction, which is what makes the cross-detector comparison well-defined. This is a deliberate departure from selecting targets at extreme clean entropy. Extreme selection inflates the clean detector AUROC toward 1.0 by construction, because such an AUROC is the selection rule reflected back, and it would overstate any measured degradation. We therefore report degradation against the clean AUROC of the *fair* pool, not the extreme one.

Measurement headroom, and why the two directions are not comparable in nats. Discrete semantic entropy over N sampled answers is bounded on both sides, $H \in [0, \log N]$: the upper bound is attained when every sample forms its own cluster, the lower when they all collapse into one. (The bound is shared with Farquhar’s Eq. (5), as set out above; the lattice structure invoked below is not.) At the deployed $N=10$ the ceiling is $\log 10 = 2.303$ nats. The two attack directions therefore face different amounts of room to move. Measured on the *score-independent fair pool* (the population the detector is characterised on, not the attacked subset), correctly-answered questions have clean mean entropy 1.380, leaving 0.923 nats to the ceiling, while wrongly-answered questions sit at 1.843, leaving 1.843 nats to the floor: the false-alarm direction has almost exactly *half* the headroom of the hide direction. The consequence is visible in the attacked cells. Of the 80 false-alarm targets, 42 finish *exactly* at the ceiling and 8 began there (so no upward move was ever possible), and a further 13 began one lattice point below it, where the total remaining headroom is 0.139 nats. That headroom is less than the success criterion’s own δ , as we set out under that criterion below. Hide targets, by contrast, reach their floor only rarely. We deliberately attach no count to that last statement here, and the reason has changed since we first wrote it. The hide cell has closed at its planned 80, so no figure we quote for it would go stale between drafts any more; what remains true is that this paper reports no hide-direction result, so a count dropped in beside the false-alarm headroom would invite an arm-to-arm reading the campaign is not powered to support. The hide-side counts belong with the hide-side analysis.

We label the population of every quantity we report. The two populations are *nested*, not disjoint, and being exact about how they are nested is what the labelling is for. Selection shuffles each correctness stratum under a fixed seed and takes a prefix: the fair pool takes the first 200 of each stratum, and each attack direction takes a shorter prefix of one of them. The 80 false-alarm targets are therefore *literally* 80 of the fair pool’s 200 correct items, and the hide targets are in the same way

a prefix of its 200 hallucinating ones. No attacked target lies outside the fair pool, and the two are not different sets of questions.

What disqualifies the attacked subset from characterising the detector is accordingly not that it is a different population but that it is a *selected sub-sample of a single stratum*: each direction draws from one correctness class only, and it is the sample the optimiser was actually run on. A quantity computed there describes correct answers, or hallucinating ones, under the attack’s own sampling. It never describes the correct-versus-hallucinating contrast that “the detector’s AUROC” names. The ceiling and granularity counts are *within*-stratum frequencies among correct targets, so we report them on the attacked pool and label them as such, while every correct-versus-hallucinating claim is made on the fair pool.

One consequence of the nesting cuts against a contrast we previously drew, and we state it rather than leave it implicit. Because the attacked targets are a score-independent sub-sample of the very same strata, a clean separation measured on them estimates the *same* quantity as the fair pool’s, only far less precisely. A discrepancy between the two would therefore be sampling error over far fewer ranked pairs, not evidence that the detector behaves differently on attacked targets. We deliberately attach no number to that discrepancy here, for the same reason we attach none to the hide-side counts above: any correct-versus-hallucinating quantity computed on the attack campaign has to pool the false-alarm arm with the hide arm, and the pooled figure would be a far less precise estimate of a quantity the fair pool already measures well. We therefore quote the fair pool’s 0.704 [0.653, 0.753] whenever we characterise the detector, we do not present an attacked-subset figure as a rival measurement of it, and we state the cross-stratum comparison with the hide-side analysis rather than here.

Two things follow, and we apply them throughout. First, false-alarm effect sizes measured in nats are *censored*: on a saturated target the attack and any other method that also saturates are indistinguishable, so raw nats understate a strong attack and overstate the similarity between methods. We therefore report the saturation rate alongside every false-alarm effect size, give effect sizes on the uncensored subset, and lead with rank-based statistics, which are invariant to monotone transformations and hence unaffected by the ceiling except through ties (handled explicitly below). Second, and more simply, *we do not compare the magnitudes of the two directions in nats at all*. A larger number of nats in the hide direction would partly reflect its larger headroom rather than a stronger attack. Cross-direction comparisons are made only via headroom fraction, rank statistics, or operating-point flips. The bidirectional formulation is unaffected; only the currency of comparison is.

Success criterion. An attack on a question succeeds only if all three conditions hold: (i) the paraphrase q' passes the bidirectional-NLI equivalence gate; (ii) the detector score moves in the intended direction by at least δ nats past its clean value ($\delta=0.25$; the cutoff is a convention, so we also report the success-rate curve over a sweep of δ rather than letting one value carry the result); and (iii) the model’s hallucination *status* is unchanged under q' , meaning still wrong for the hide attack and still correct for the false-alarm attack. We check that status with the same greedy decode and span oracle used to label the pool. Condition (iii) is essential: a score can fall because the paraphrase accidentally elicited a *correct* answer (removing the hallucination rather than hiding it), or rise because it elicited a wrong one; without the status check these would be miscounted as attacks. We report the status-gated success rate alongside the weaker entropy-only rate so the attrition between them is explicit.

A quarter of the false-alarm denominator cannot satisfy condition (ii) at any usable δ . Condition (ii) interacts with the discreteness of the score in a way we disclose here, where the criterion is defined. The interaction does not merely make some targets hard. It makes them impossible, while leaving them in the denominator of every rate we report. At $N=10$ the discrete plug-in entropy is not just bounded by $\ln 10$; it is confined to the 39 values obtainable as the Shannon entropy of an integer partition of 10 into cluster sizes, and the top two of those, 2.164 and $\ln 10 = 2.303$, lie 0.139 nats apart. A target whose clean score is already 2.164 thus has 0.139 nats of headroom *in total*: no paraphrase whatsoever, adversarial or benign, can move it $\delta=0.25$ nats upward, and a target already at $\ln 10$ cannot be moved upward at all. Those two points hold 21 of the 80 false-alarm targets (8 at the

ceiling, 13 one point below), so 26.2% [17.9, 36.8] of the pool is unwinnable by construction. The interval is Wilson. At $\delta=0.25$ the unwinnable set coincides exactly with the top-decile set, because the next lattice point down, 2.025, still leaves 0.277 nats. The δ -sweep we offer above does not rescue these targets: it re-prices the cutoff, not the headroom. Every $\delta \geq 0.139$ is unsatisfiable for all 21 of them, and no $\delta > 0$ whatsoever is satisfiable for the 8 already at the cap. Recovering even the 13 would need a cutoff below 0.139 nats, which is finer than one step of the lattice near its top. We therefore report this count alongside every false-alarm success rate, so that a rate is read against the part of its denominator that could in principle have produced a success. This is a property of the discrete estimator at the deployed sample budget, not of our attack: it applies identically to any method evaluated with a fixed- δ criterion on this score. It is also the sharpest of the claims that does *not* carry to the likelihood-weighted estimator, whose real-valued weights leave no lattice for a δ to fall between.

Evaluation and reporting. We report every rate with a bootstrap 95% confidence interval over questions. Beyond the aggregate score shift we report operating-point flips: fixing a detector threshold at a target false-positive rate on clean data, we count hide targets that move from flagged to unflagged and false-alarm targets that move from unflagged to flagged. Those are the operationally meaningful events. Fixing that threshold is less innocent than it sounds, and we do it explicitly rather than by quantile. Because the score is discrete, a rule that flags when $H \geq \tau$ realises only the false-positive rates attained by some clean negative, so a nominal target such as 5% or 10% generally does not exist; interpolating between order statistics returns a threshold that silently runs at a different rate, in our case answering a 10% request with a threshold that operates at more than twice it. We therefore select τ as the smallest threshold whose *achieved* clean false-positive rate is at or below the stated budget, assert that achieved rate at run time, and report it rather than the nominal one. Enumerating the achievable rates this way is what turns the ceiling into an operating constraint (Section 5). We additionally break out the subset of would-be successes voided by condition (iii) because the model’s answer flipped, since these are candidates for equivalence-gate leakage and *lower*-bound how often the gate admits a non-equivalent paraphrase, which is an *upper* bound on the fidelity of the NLI constraint and not a lower one (Section 6 states it the same way). Both detectors are evaluated with a fixed sampling seed, so per-question entropy is reproducible and the SE–SRE comparison is like-for-like rather than one deterministic detector against one stochastic one. The operating-point FPR sweep and the entropy-cutoff curve are exploratory sensitivity analyses, not independent hypothesis tests; the confirmatory comparisons are stated in advance and the AUROC degradation is reported with a paired bootstrap confidence interval that resamples questions jointly.

Independent-clusterer attribution. Semantic entropy is defined by a DeBERTa-large-MNLI clustering of the sampled answers, and our attack’s feasibility gate uses the same model; a measured entropy move under this clusterer therefore cannot be attributed to a change in the model’s answer distribution rather than to the clusterer’s own inconsistency. To separate these, we re-cluster the identical answer samples under equivalence relations independent of the DeBERTa-MNLI supervision: an exact-match relation (canonical-normalised string equality) and an embedding-cosine relation using e5-base-unsupervised, the E5 checkpoint trained only by contrastive pretraining before any NLI fine-tuning. We report the attack’s mean *intended move* under each as a bracket, with the shared-NLI arm as the confounded upper bound and exact-match as the saturation-limited lower bound. The bracket is a statement about the intended move only, and it does not survive differencing. Every budget-controlled quantity we report subtracts an arm-specific benign comparator, and a lower bound on the attack’s move together with a lower bound on the benign comparator is not a lower bound on their difference. Exact-match is the strictest relation, so it splits the most, so its clean baselines sit closest to the $\log N$ ceiling, and that same saturation suppresses the benign best-of- m by more than it suppresses the attack’s single unselected score. The adjudicator therefore lies outside the bracket on the paired net (Section 4), and we report where it does rather than let the column headers imply otherwise. We find that no cheap independent oracle adjudicates cleanly: exact-match saturates on short factoid answers, and the embedding oracle, though excellent on easy paraphrase pairs (STS-B AUROC 0.99), is near-chance on the word-preserving, meaning-shifted pairs the attack produces

(hard-negative AUROC 0.51), so its clustering cannot be trusted to adjudicate. We therefore build and validate a victim- and NLI-independent LLM-judge equivalence oracle (Qwen2.5-7B-Instruct). In the deployed symmetric configuration, on the same labelled pairs, it reaches hard-negative accuracy 0.93 [0.90, 0.96] ($n=300$) where the embedder is near-chance (0.51). These strata are clean gold-alias pairs, a domain-matched *proxy* for the deployed clustering decision over sampled answers, and a validation on messy real sample pairs remains owed (Limitations). *We disclose a deviation.* Our pre-registered acceptance gate required both hard-negative *and* positive-alias recognition above 0.8; the judge clears the first but not the second (0.65 [0.59, 0.70]): on those clean pairs it over-splits some genuine aliases, in part correctly rejecting noisy TriviaQA alias labels. Over-splitting inflates baseline entropy, which would make the judge *conservative* for the false-alarm direction (a surviving effect is understated) but *not* for hide; we therefore re-scoped it to false-alarm adjudication under a hard-negative-primary criterion. This is an outcome-triggered re-specification of a pre-registered gate, which we flag rather than bury, and we report the NLI and exact-match bracket *alongside* the judge so the judge’s intended move can be read against two relations that do not share its supervision. We claim no more for that than it delivers. The bracket covers the intended move and not the budget-controlled comparisons the judge’s conclusions are drawn from, and on the ones the paper leads with, the paired nets against a selected benign maximum, the judge lies outside it; no independent arm bounds those quantities, so on exactly them the re-scoped oracle does stand alone, and we state that rather than let the bracket stand in for a guarantee it does not give. *We flag one further thing, because the argument for the re-scope did not survive the data it was made about.* The judge was validated on clean alias pairs, but what it actually clusters is sampled answers. On those it under-splits rather than over-splits. It merges all ten samples into one cluster on 27 of the 80 false-alarm targets’ clean baselines, against the NLI clusterer’s 4. The re-scoping decision stands, since it rests on the hard-negative accuracy the judge does clear, but the conservativeness rationale attached to it does not, and the direction of the residual bias is anti-conservative rather than conservative; Section 6 states what that costs and what it leaves owed. Re-clustering the answer samples under this validated judge has so far been exercised only on a preliminary $n=6$ machinery-validation pass, which establishes that the judge arm runs end-to-end and reports correctly, and nothing else. We quote no point estimate from it, for three independent reasons. The net confidence interval includes zero at that sample size, so the effect is not claimed. The pass was specified in advance as a pipeline check, and its own record marks its figures as not for publication. And the per-target diagnostics that would otherwise invite an arm-to-arm reading are, at $n=6$, too coarse to support one. The last reason is worth stating exactly, because it is a resolution limit rather than a bound: the arm-level diagnostic is a mean over the six targets of each target’s rank within its own benign-move distribution, and a target contributes at most $K=5$ benign draws, so the arm-level statistic moves in steps of roughly $1/30$. The difference between the shared-NLI and the judge arm is about one such step. A gap that small reflects the sample size, not a measured ordering of the arms. Which arm the judge’s verdict resembles was therefore left to the confirmatory run under the validated judge, which was the gate on that question as much as on the effect itself and which completed on 2026-08-29. The statistic reported there is the randomised-tie exceedance test described in Section 4, which prices the attacker’s search budget into the null analytically; the percentile-in-benign indicator is shown as a diagnostic only, being upward-biased by that same budget. Absent a cleared claim, attribution is an explicit bracket, not an assertion.

The pre-registered sample-size threshold was met by the campaign and missed by the test, and we state that here rather than only in Limitations. The pre-registration set $n \geq 80$ with $m=50$ benign paraphrases per target. The attacked false-alarm stratum reached its 80 targets, and nothing else in the analysis did: on three of them the feasibility gate rejected all 250 benign attempts, so they have no benign arm at all and are excluded rather than absorbed, leaving the exceedance test and every paired quantity at $n=77$; of those 77, the full $m=50$ budget was achieved on 71, the remaining six running at m of 1, 26, 27, 27, 35 and 38 because attempts are capped at five times the declared budget and the same gate rejects the rest. The convention-sensitivity row that excludes no-op targets runs on 69. A short benign arm costs the exceedance test power rather than calibration, since its null carries each target’s own m ; the threshold was nonetheless not met by the statistic the decision rule is stated on, and a non-rejection has to be read against that.

Transfer and defense. To test whether the vulnerability is paradigm-level rather than specific to one implementation, we take a paraphrase optimized against one detector and measure the score change it induces on another ($SE \leftrightarrow SRE$), and, as a stretch, on a hidden-state probe (SEP [11]) that never samples. As a first defense we average the detector score over d paraphrases of whatever input is received (d distinct from SRE’s samples-per-reformulation K and the benign-floor budget m), on the intuition that a single adversarial phrasing is diluted by its own paraphrases; this mirrors SRE’s built-in reformulation and lets us ask whether that mechanism confers attack resistance.

4 Experiments

Setup. The victim model is Llama-3.1-8B-Instruct in 4-bit quantisation on a single 16 GB GPU. Semantic entropy draws $N=10$ answers at temperature 1.0 and clusters them by bidirectional DeBERTa-large-MNLI entailment; semantic reformulation entropy uses $R=3$ reformulations, $K=8$, temperature 0.8. Our SE replication on TriviaQA does not have one AUROC: on a single run of 2000 questions, the number depends on which correctness convention labels it, and the three we computed span 0.096. All three are adjudicated by the same alias-aware span oracle, so the convention is the only thing that varies across them. Under the greedy label used everywhere else in this paper the replication scores 0.694; under a majority-of-samples label 0.729; and under an all-samples-correct label it reaches 0.790, against the published SE figure of 0.828. We lead with the operative 0.694, because a replication figure obtained under a label convention the paper does not otherwise use is not a check on the pipeline the paper actually runs. That choice puts the gap to the published figure at 0.134 rather than 0.038, and we do not attribute it: the NLI backend differs from the deberta-v2-xlarge of some prior work (Limitations), but so does the correctness convention, and we have not run the ablation that would separate the two.

We flag the all-samples label specifically, because it is the one convention mechanically coupled to the score. It calls a question correct *iff* all ten sampled answers are correct, and semantic entropy is the entropy of the clustering of those same ten samples. An all-correct sample set is far likelier to be semantically homogeneous, hence low-entropy, so part of the AUROC it awards is the score being scored against itself. Validating a paper whose third contribution is score-independent selection on the most score-entangled label available would undercut the contribution. Reporting the operative figure also resolves what would otherwise be an unexplained tension with the next sentence: targets are drawn by the stratified score-independent sampler, and on the resulting fair pool the clean detector scores AUROC 0.704 [0.653, 0.753], against the 1.0 that score-dependent selection manufactures by construction. That interval contains the operative replication figure but excludes the all-samples one. That 0.704 is one of the figures Section 6 traces to a single June 2026 generation cache, whose sampling configuration is under-recorded and against which the later $N=40$ run shows a measured generation drift; neither has been bounded for this number.

Reading the null control. Every cell of Table 1 is measured on the *false-alarm stratum of the attacked pool*, the 80 score-independently selected correct-answer targets the optimiser was run on. No cell is measured on the fair pool used to characterise the clean detector (Section 3). The two are nested: these targets are a prefix of the fair pool’s 200 correct answers. The distinction is therefore not between disjoint question sets but one of scope, a single correctness stratum under the optimiser’s own selection. No number is carried between them. Table 1 places the optimised attack against two null bands, under each clustering relation the campaign instantiates. The first band is a same-question *seed-noise* floor. The second is a *benign-paraphrase* floor of $m=50$ NLI-passing non-adversarial rephrasings, and its comparison against the attacker’s ~ 180 -candidate (~ 181 -evaluation) search is budget-matched *analytically* by the exceedance test below rather than by drawing m equal to that budget, subject to the budget parameter that test needs being only partly identified from what the run recorded (below). (m is distinct from SRE’s samples-per-reformulation K .) The shared-NLI arm is the detector’s own clusterer and is the *confounded upper bound*: it cannot separate a real change in the answer distribution from the NLI model’s own inconsistency, because that same model gates the attack’s paraphrases. Exact-match is the *saturation-limited lower bound*: it is the strictest

Table 1: Null-controlled attack effect, SE / false-alarm direction, per clustering relation, on the false-alarm stratum of the attacked pool. Denominators differ by block; see the text, which also says why the embedding relation has no column. The bound labels in the column heads describe the attack’s own mean intended move and no other row: every comparison below it subtracts an arm-specific benign comparator, and the bracket does not survive that subtraction (Section 3). *The test clamps each target’s measured tie multiplicity b_j to A , since b_j counts attack candidates at the maximum. The clamp fires on no target at $A=181$, on 2 at $A=121$ and on 22 of the 77 (29%) at $A=41$, so that last row overwrites the measured multiplicity on 29% of them. Clamping b_j down enlarges each tied benign draw’s credit $1/(b_j+1)$ and so raises p : the direction is conservative, and it is the only reason the exact-match and adjudicator totals drift up, by 7 and 2, across the range.

	Shared NLI (confounded UB)	Exact-match (saturating LB)	LLM-judge (adjudicator)
<i>Bands: mean intended move (nats) [95% paired bootstrap over targets]</i>			
Attack ($n=80$)	+0.526 [+0.414, +0.642]	+0.213 [+0.127, +0.311]	+0.291 [+0.152, +0.440]
Seed-noise floor ($n=80$)	−0.015 [−0.097, +0.070]	+0.026 [−0.020, +0.074]	−0.074 [−0.151, +0.004]
Benign floor, mean draw ($n=77$)	−0.049 [−0.166, +0.066]	+0.002 [−0.096, +0.097]	−0.071 [−0.172, +0.028]
Benign floor, best of m ($n=77$)	+0.363 [+0.232, +0.490]	+0.213 [+0.117, +0.313]	+0.422 [+0.299, +0.546]
<i>Claim statistic: randomised-tie exceedance test, $n=77$, at the budget upper bound $A=181$ unless a row says otherwise</i>			
Exceedance p, median of 101 tie draws	0.939	1.000	1.000
range over those 101 draws	[0.229, 1.000]	[1.000, 1.000]	[1.000, 1.000]
draws reaching $p \leq 0.05$	0/101	0/101	0/101
median p at $A=121$ (point estimate)	0.403	1.000	1.000
median p at $A=41$ (lower bound)*	< 0.001	1.000	1.000
Exceedances S , seed-0 draw / expected	30 / 20.4	383 / 20.4	636 / 20.4
median S over those 101 draws	28	383	652
Effective benign budget n_{eff}	122.5	8.7	4.8
<i>Ceiling saturation over the false-alarm targets, and the two inadmissible tie rules</i>			
Saturation at log N : clean / attacked	10.0% / 52.5%	46.2% / 70.0%	1.2% / 10.0%
Power against a ceiling-saturating attack	0.56	0.00	1.00
Diagnostic: strict-tie p	< 0.001	1.000	1.000
Diagnostic: conservative-tie p	1.000	1.000	1.000
<i>Supplementary: paired net under convention (d), $n=77$, against three comparators</i>			
vs. best of m on this arm	+0.110 [+0.063, +0.161]	−0.027 [−0.064, +0.015]	−0.173 [−0.302, −0.039]
vs. the NLI-selected benign candidate	+0.110 [+0.063, +0.161]	+0.018 [−0.021, +0.064]	+0.016 [−0.112, +0.150]
vs. the mean benign draw	+0.381 [+0.303, +0.464]	+0.101 [+0.050, +0.159]	+0.175 [+0.045, +0.312]
Attack’s mid-rank percentile in benign	0.931 [0.903, 0.955]	0.736 [0.686, 0.785]	0.700 [0.637, 0.761]

relation, so it splits the most and saturates on short factoid answers. Its bracket with the NLI arm holds on the attack’s own mean intended move, the first row of Table 1, and it does not survive differencing against a benign comparator (Section 3), so it is not a bound on any other row of that table. The e5-base-unsupervised embedding relation is NLI-supervision-free but near-chance on the adversarial, word-preserving case (hard-negative AUROC 0.51), so it cannot adjudicate. It is also *absent* from the campaign reported below, and we state that rather than leave a blank column to be misread: the definitive run was launched with an empty embedding-model setting, so the embedding arm’s benign list has length 0 on all 80 targets of the false-alarm stratum and its attack move is undefined. There is no embedding measurement here. Its absence is not a measured null, and we do not report it as one. The LLM-judge (Qwen2.5-7B-Instruct) is the **adjudicator** for the false-alarm direction, under the disclosed re-scoped gate (Methods). It is victim- and NLI-independent, and it is validated to 0.93 [0.90, 0.96] ($n=300$) on domain-matched gold-alias hard negatives, which are a proxy for the deployed clustering decision.

The bands themselves are the mean intended move (nats) with a 95% paired bootstrap over targets, and they are descriptive only: the band ordering is not the test. The test is the randomised-tie exceedance statistic developed next, and ceiling saturation is reported beside that statistic because saturation governs how far the nats bands can be trusted.

The claim statistic, and why the obvious ones fail. The attack spends 181 objective evaluations per target, the original question and the 180 candidate paraphrases it proposes (Section 3), and reports the *maximum* over whichever of those its feasibility gate admits. Any comparison against individual benign draws is therefore upward-biased by construction: under the null a max-of-181 sits at $\sim 99\%$ of an individual-draw distribution, not 50%. We therefore price the search budget into the null analytically. Under the hypothesis that the optimiser carries no signal, a target’s A candidates and its m benign draws are exchangeable, so $F(\text{attack-max}) \sim \text{Beta}(A, 1)$ and the number of benign draws above the attack’s maximum is $K \sim \text{BetaBinomial}(m; 1, A)$, with $\mathbb{E}[K] = m/(A+1)$. Here A is the attacker’s search budget, meaning the number of candidates the reported maximum actually ranges over. A is distinct from the per-question sample budget N , which is 10 throughout this section. N is not 10 throughout the paper: Sections 5 and 6 characterise the clean detector on the fair pool at $N=20$ and $N=40$ as well. Fewer exceedances than that expectation is evidence for the attack; the one-sided p -value follows by exact convolution over targets, and the effect size is the *effective benign budget* $n_{\text{eff}} = m/\bar{K} - 1$, read as “the search is worth n_{eff} random paraphrases”.

Ties are not a technicality here. Discrete semantic entropy is bounded above by $\log N$ (Section 3), and benign paraphrases reach that ceiling too, so the score has an *atom* where the exchangeability null assumes a continuum. We resolve ties by **randomised (exchangeable) breaking**: a benign draw tied with the attack’s maximum outranks it with probability $1/(b+1)$, where b is the number of attack candidates also achieving that maximum. Simulation shows why the alternatives are inadmissible, and the two simulations it draws on have to be read separately, because they differ in how the tie multiplicity b is generated. At full saturation ($q=0.05$, $m=30$), in the simulation that *redraws* b from its null distribution, counting ties strictly gives a null rejection rate of 0.995 and counting them against the attack gives power 0.05 at a two-fold effect; in that same simulation the randomised rule’s analytic-null level is 0.051. In the deployment-faithful simulation, which takes b as *measured* on the attack’s own candidates, the randomised rule stays calibrated at that cell, with analytic-null level 0.021. The measured- b figure is the one that bears on the deployed test, because b is measured in deployment; redrawing it runs b about one higher, which shrinks the $1/(b+1)$ tie credit and pushes the level up.

Under the empirical $n=80$ headroom distribution, the randomised rule against that analytic null reaches power 0.51 at an achieved level of 0.025 for $m=30$, and 0.77 at 0.041 for the pre-registered $m=50$; levels and powers are paired on the same 4,000 simulated trials per cell, and the analytic null’s own nominal tails at those cut points are 0.031 and 0.044, so the shipped test is stricter than it believes itself to be. We quote the deployed test throughout. On the same draws, a critical value calibrated against the true simulated null reaches 0.66 at level 0.056 and 0.84 at 0.064 for those two cells. Such a calibration is an oracle the deployed test does not have. The difference between the two columns is a level, not a statistic: at a common m the analytic p -value is strictly increasing in the exceedance total alone, so the deployed and oracle tests are the same cut rule on the same statistic, applied at different cut points. Re-cut to the oracle’s achieved level, the deployed test’s power equals the oracle’s to three decimals at every m we simulate. The analytic null therefore costs nothing in discriminating power and buys conservatism. Unlike the oracle, it also delivers a level the pipeline can state without knowing the truth it is testing. What it does cost is the operating point rather than the statistic: the cut we ship is the stricter one, so a non-rejection must be read against the deployed power and not the oracle’s. Both inadmissible rules are reported as diagnostics: their disagreement is a direct indicator that ceiling saturation, rather than attack strength, is driving a comparison. Note that b must be *measured* rather than estimated, since it is precisely where attack strength appears once the ceiling pins the maximum’s value; our optimiser records it per target.

The budget A is not identified, and the verdict does not rest on it. The optimiser spends exactly 181 objective evaluations on each of the 80 targets, and it is tempting to read A off that count. It is the wrong count. A candidate can become the reported maximum only if it also passes the feasibility gate, and our gate is applied *lazily*: only candidates that already match or beat the running best are submitted to it. Of the 14,400 proposed children, 4,703 (32.7%) were feasibility-checked at all, and 3,058 of those passed, a pooled pass rate of 0.650; the rest were scored, dropped on the pre-filter, and never gated. Their feasibility is not recoverable, because the run stores the feasible candidates’

Table 2: Convention sensitivity of the supplementary paired net (nats, 95% paired bootstrap, 20,000 resamples), on the same false-alarm stratum of the attacked pool. The text gives each convention, the arithmetic behind each n , and why (d) is the locked one.

Convention	n	Shared NLI	Exact-match	LLM-judge
<i>Paired net (nats) over the false-alarm targets, 95% paired bootstrap in brackets</i>				
(a) as-is	77	+0.184 [+0.114, +0.262]	+0.008 [−0.044, +0.068]	−0.119 [−0.254, +0.019]
(b) no-op <i>pairs</i> neutralised to 0	77	+0.126 [+0.072, +0.186]	−0.006 [−0.043, +0.036]	−0.136 [−0.266, −0.003]
(b') benign may decline, no-op rows only	77	+0.126 [+0.072, +0.186]	−0.014 [−0.054, +0.030]	−0.149 [−0.280, −0.016]
(c) no-op <i>targets</i> excluded	69	+0.141 [+0.081, +0.206]	−0.007 [−0.048, +0.040]	−0.152 [−0.300, −0.005]
(d) benign may decline, every row (locked)	77	+0.110 [+0.063, +0.161]	−0.027 [−0.064, +0.015]	−0.173 [−0.302, −0.039]

scores but not the ungated candidates’ strings, so no re-analysis of what we recorded can settle it. $A=181$ is therefore an *upper* bound and not a measurement. The lower bound is the case in which no never-gated candidate would have passed, leaving the maximum ranging over the recorded feasible set plus the original: a mean of 41 over the 77 targets the test runs on. Extrapolating the observed pass rate to the never-gated children puts the point estimate near 121. The defensible range is [41, 181], and we report the test across it rather than at one end of it.

What that range does to the three arms is not symmetric, and it is why the defence is stated this way. The shared-NLI arm’s p -value is *not identified*: its median over 101 tie draws runs from 0.939 at $A=181$ to below 0.001 at $A=41$, and it crosses the 5% cut near $A=90$. Nothing we recorded rules that region out. The two re-scored arms do not move at all, and the reason is not that they are insensitive: A enters this test through the null and almost nowhere else. The exceedance total S is built from the benign draws, the attack’s maximum and the measured tie multiplicity, so apart from the one clamp on that multiplicity disclosed in Table 1’s caption it is the same statistic at every budget, and what the range moves is the expectation S is weighed against. The adjudicator’s median p is 1.000 at every A in [41, 181] and at every one of the 101 tie draws, and so is the exact-match arm’s; their medians are 652 and 383 at $A=181$, unchanged at $A=121$, and 654 and 390 at $A=41$ once the clamp has fired, against an expectation rising from 20.4 to 30.4 to 88.2, so both stand more than fourfold above it at the hard lower bound. The shared-NLI median moves as little, from 28 to 31: its p -value moves because the expectation passes its total, not because the total does. **The decision rule was pre-registered on the adjudicator arm alone** (below), and that arm’s verdict is invariant to the one parameter we cannot pin down. That invariance, and not any headroom in the shared-NLI arm, is what the non-rejection rests on. Reporting the NLI arm’s 0.939 beside it is reporting a p -value at the end of a range most favourable to the conclusion we draw, and we say so rather than quote it as though the budget behind it had been measured.

Decision rule (single, pre-registered, with disclosed deviations). Reframe (a) is the claim that “the attack targets the detector”. It is claimed iff, at $n \geq 80$ with $m=50$ benign paraphrases per target, the adjudicator clusterer’s randomised-tie exceedance test rejects at the 5% level in the direction of the attack. That design was fixed in advance at a simulated power of 0.84 against a two-fold effect and 0.99 against a three-fold one, chosen over a cheaper $m=30$ (power 0.67 at the design-time simulation settings, 0.66 on the higher-precision re-run quoted above) because our own selection-on-noise measurement indicates the true effect is modest and the design must be able to see a small one. We disclose that those pre-registration figures were the oracle-calibrated ones; the deployed analytic-null test, cutting at its stricter achieved level, delivers 0.77 at $m=50$ and 0.51 at $m=30$. That shortfall is a level rather than a weaker statistic, since re-cut to the oracle’s level the two agree to three decimals, as above. It is nonetheless the operating point we actually ship. The ordering that drove the choice is unchanged and we do not re-choose m after the fact, but the design rejects less readily than the number we picked it on, and a non-rejection must be read against 0.77 rather than 0.84. If the test does not reject, (a) is not claimed, and the NLI/exact-match bracket together with the judge’s point estimate is reported as the honest result. We are precise about what is bounded there, because the pre-registration was not: the bracket bounds the mean intended move, and the judge’s point estimate lies outside it on the paired nets of Table 1 that the closing paragraphs lead with, so the word *bounded*

is not available for the quantities this section concludes from. That is the branch the run took, and the paragraphs after the deviations report it.

Four deviations from the original pre-registration are disclosed as outcome-triggered rather than folded silently into “the pre-registered rule”. The adjudicator moved from the sentence embedder to the LLM-judge, triggered by the embedder’s 0.51 hard-negative AUROC. The benign floor moved from individual draws to a budget-corrected comparison, triggered by the winner’s-curse mismatch surfaced in adversarial review. The budget correction then moved from brute-force matching to the analytic exceedance null, triggered by the matched design pricing out at what we then costed as 228 GPU-hours. We correct that figure here rather than leave it standing: re-derived from the wall clock of runs that have since executed, the matched design is about 99 GPU-hours. The deviation still holds, because an analytic null that costs nothing is preferable to a four-day matched arm. Even so, the trigger we recorded was $2.3\times$ too large, and a reader checking whether the rule was tuned is entitled to the corrected number. And the tie rule moved from conservative to randomised breaking, triggered by the simulation above showing the conservative rule has no power under the ceiling. Each trigger is objective, each was recorded before the corresponding data was seen, and the full sequence is given here rather than in an appendix precisely because a reader is entitled to check that the rule was not tuned to the result.

Result: what the rule returns. The confirmatory run completed on 2026-08-29. It attacks the *false-alarm stratum of the attacked pool*, all 80 score-independently selected correct-answer targets, which is the pre-registered $n\geq 80$ met by this population and by no other quantity in this section. The run scores each target’s outcome under three clustering relations, against $m=50$ benign feasible paraphrases drawn from the same proposer and passed through the same feasibility gate, plus a three-seed same-question band. The judge is Qwen2.5-7B-Instruct. **The rule returns a non-rejection, and it does so in every arm.** In the adjudicator arm the randomised-tie exceedance test gives a median p of 1.000 across 101 tie-break realisations, with no realisation anywhere near the 5% cut and no admissible budget that moves it; the shared-NLI and exact-match arms give 0.939 and 1.000 at the declared $A=181$. Reframe (a) is therefore **not claimed**. The direction is worth stating, but it has to be stated arm by arm rather than for the design as a whole, because the expectation it is measured against moves with the budget and the budget is not identified. The observed total is itself a tie-break realisation, and we quote it as one rather than setting it beside a median p as though it were fixed. At the seed-0 draw it is 30, 383 and 636 against an expectation of 20.4 at $A=181$. The medians over the same 101 draws that the p -values summarise are 28, 383 and 652. The exact-match and adjudicator totals sit *above* their null expectation at every admissible budget, which is the direction *against* the attack and is the invariance the verdict rests on. The shared-NLI total does not: it is above expectation at $A=181$, but the expectation catches it near $A=131$ and passes it below that, so over most of the defensible range the NLI statistic sits on the attack’s side of its own null, and below $A\approx 90$ a majority of tie draws would reject. That is a statement about a parameter we cannot measure rather than a result, and it is the reason the rule is stated on the arm that does not depend on it. The shared-NLI total is also the one that moves across tie draws, over [16, 41], and 0.229 is its lowest p at $A=181$ and not in general. In the arm the optimiser actually searched, its search behaves like $n_{\text{eff}}\approx 122$ random feasible paraphrases. We do not read that as fewer than the budget it spent: n_{eff} is denominated in feasible paraphrases while the 181 counts scored evaluations, and against the feasible budget’s own point estimate of 121 it falls short of nothing. Off that arm the same effect size collapses: n_{eff} is 8.7 under exact-match and 4.8 under the pre-registered adjudicator, so a search costing 181 evaluations ranks where the best of about five random benign paraphrases would. That is not a clean measure of the search’s strength, since off the objective arm the attack’s query is re-scored rather than selected and so inherits the mis-specification set out below, but it is the effect size the pre-registered arm returns and it points the same way as that arm’s p . Doubling the measured tie multiplicity, the one perturbation that moves p downward, leaves the NLI arm at 0.229 and the other two at 1.000.

A non-rejection is not a measurement of zero, and we do not report it as one. At the deployed operating point this design has power 0.77 against a two-fold effect at $m=50$, so the correct reading is that an effect of that size is *not detectable at this n on this population*, not that none exists. The same discipline governs every supplementary interval in this section, and we state it once as a rule so

that it can be checked against every later sentence: an interval that covers zero is reported as *covering zero*, never as null. The exact-match net covers zero. So does the index-matched adjudicator net of the last paragraph below, and it gets the same wording rather than a stronger one.

Denominators, and the convention, both disclosed. Four different n appear above, all of them inside the false-alarm stratum of the attacked pool, and none of them may be substituted for another. The campaign is 80; every benign-referenced statistic runs on 77, because three targets have no benign arm at all; convention (c) runs on 69; and the full $m=50$ budget was achieved on 71. Three of those four are row denominators in Table 1, and the table is read block by block: the bands run over all 80 attacked targets, every benign-referenced row over the 77 with a non-empty benign arm, and the convention (c) row of Table 2 over 69. The exceedance test and the paired net are both benign-referenced, so both run on 77. The three empty arms (qb_565, qz_3393, qb_2689) are the targets whose every one of 250 feasibility attempts was rejected, and on each of them the optimiser also found no feasible paraphrase. Taken together, those two failures are evidence that no feasible paraphrase of those questions exists, rather than that the sampler was unlucky. That second property is not peculiar to those three: 11 of the 80 targets have no feasible candidate at the optimiser’s best, identically identified by three independent criteria (the returned query equals the original, the optimiser records no improvement, and its feasible-at-best count is zero). What singles out these three is that the benign sampler failed on them too, and it is the empty benign arm, not the optimiser’s no-op, that removes a target from the denominator. They are excluded and stated, never absorbed into a denominator of 80. Convention (c) of Table 2 drops eight rows, not eleven, and the arithmetic is worth spelling out because $77-11$ is not 69: three of the 11 no-op targets are the three with an empty benign arm, and those three are already outside the 77, so only the remaining eight are excluded there. Those no-op targets bear on the paired net for a reason that is structural rather than statistical. The optimiser returns the original question when nothing beats it, so the attack’s move is exactly 0 on the 11 and, in the NLI arm it was optimised against, is bounded below by 0 everywhere (measured minimum +0.0000 over all 80). The feasibility gate forbids the benign arm from drawing the no-op, so its maximum is unconstrained and is negative on 10 of 77 targets under NLI. The paired net therefore credits the attack with the null’s losses unless a convention intervenes. Six further arms are short, at m of 1, 26, 27, 27, 35 and 38. The resulting inflation of the net is +0.006 nats as-is and +0.001 under the locked convention, because the benign score distributions are lumpy: the discrete estimator has only 39 attainable values at $N=10$, and 29% of a target’s benign NLI draws already sit at that target’s own maximum, so a maximum over 26 draws almost always equals the maximum over 50. That is a disclosure, not a correction. The two disclosures pull the same way and neither is large enough to move a conclusion: the short arms inflate the net, and the locked convention deflates it. Finally, and non-negotiably: convention (d) of Table 2 was fixed in a pre-registration written when 68 of the 80 targets were visible, which locked it in advance of the last 12 and after the effect of every candidate convention on every arm had been computed. It is not a blind pre-registration and must not be read as one. What can be said for it is stated in that document and repeated here: it is the convention least favourable to our own attack in all three arms, which is the direction that makes a post-hoc choice non-self-serving, and all four are reported, as that document requires, so that a reader can price the choice rather than take it on trust. Of the alternatives, (a) is indefensible on principle and (c) changes the denominator.

Where the exceedance null is, and is not, correctly specified. The attack’s objective is the shared-NLI entropy and nothing else, so the query it reports is an argmax over that target’s feasible NLI candidates, at most 181 of them, and the exact-match and judge values are independent re-scorings of that single query. The exceedance null prices in a maximum over A exchangeable draws. That is exactly what the NLI arm contains and exactly what the other two do not: there the attack contributes one unselected observation, so the null expects it to be far more extreme than it has any reason to be, and p is inflated. The mis-specification runs against the attack, which is the conservative direction for the claim we are not making, but it has a consequence a reader should not have to derive: the exact-match and judge non-rejections are weaker evidence than the NLI one, and the NLI arm is the only place where this test is asking the question it was designed to ask. That holds even though the

NLI arm is itself confounded as an effect estimate. That is not the same as resting the verdict on the weakest arm, and a reader who joins this paragraph to the budget argument above will otherwise conclude that it is. The verdict is the withholding of a claim rather than a claim, so a mis-specification that inflates p can leave reframe (a) unclaimed but can never manufacture it; what the adjudicator’s invariance across [41, 181] buys is procedural, that the pre-registered rule returns one answer at every budget we cannot rule out, while the evidence *about* the attack lives in the NLI arm, where the honest report is that the p -value is not identified.

For one of the three arms that weakness is total, and we state it rather than let three reported p -values read as three confirmations. The deployed cut is $S \leq 11$: exact convolution of the per-target nulls gives $\Pr(S \leq 11) = 0.0296$ and $\Pr(S \leq 12) = 0.0501$, so 11 is the largest total this test can reject on at $n=77$. The exact-match arm cannot produce one. Its benign draws are pinned at the $\log N$ ceiling on 1,461 of 3,704 draws (39.4%), against 10.7% of the NLI arm’s draws and 1.2% of the judge’s; that is a draw-level rate, not the target-level clean saturation Table 1 reports. Every one of those ties hands the benign side tie credit. Simulating the strongest alternative this design exists to catch, an attack that reaches the ceiling on every target at that target’s measured tie multiplicity, the exact-match total has expectation 150 and its smallest value over 20,000 draws is about 110. Its rejection region is unreachable. Power against that alternative is 0.56 in the shared-NLI arm and 1.00 in the adjudicator’s, and 0.00 in exact-match. The exact-match $p=1.000$ is therefore a certainty of the arithmetic rather than evidence about the attack: the non-rejection rests on two arms that could have rejected and did not, not on three.

The shared-NLI arm carries a mis-specification of its own, and it runs the other way, so we disclose it here rather than leave it to the pre-registration. Exchangeability requires the attack’s reported value to be a draw from the same feasible-paraphrase distribution as the benign draws. On a no-op target it is not: the optimiser returned the original question, so the attack’s move is a structural 0, while the feasibility gate rejects an identical “paraphrase” as its first test, so the benign arm can never produce one. The eight no-op targets that have a benign arm therefore donate an observed exceedance count of exactly 0 against an expectation of 1.93, and they do so in all 101 tie-break realisations rather than on average. That pushes S down and p toward rejection *in the attack’s favour*. It is an anti-conservative violation of the one arm the test is correctly budgeted for, it is unresolved, and at a borderline result it would matter. It does not change this one: with 1.93 of the shared-NLI arm’s 20.4 expected exceedances handed to the attack for free, the test still does not reject at any realisation. The other two arms show the reverse, since their no-op targets donate far *more* exceedances than expected. Nothing about this violation is therefore shared across the three.

Nor does the opposite correction rescue the attack. Pricing those arms at a budget of one would over-correct: the attack’s judge value is not an independent draw but the judge score of an NLI argmax, so its effective budget is whatever the argmax inherits through the correlation between the two clusterers, somewhere strictly between 1 and A . We do not model that quantity. The index-matched row of Table 1 sidesteps the need to model it, by holding the selection rule fixed across both sides, and it covers zero.

What the judge arm does and does not say. The paired net is **supplementary** to the exceedance test. Table 1 reports it under the pre-registered convention (d) (Table 2) and against three comparators, because its sign depends on which one is used. Under *every* convention the judge column is measured against a benign maximum selected on the judge while the attack’s judge score is unselected; that asymmetry is what puts the judge column below zero, and the convention is not. Under the locked convention the paired net in the adjudicator arm is -0.173 [$-0.302, -0.039$] nats, an interval lying entirely below zero, and we state the sign plainly: measured this way, the optimised attack moves the adjudicator’s semantic entropy *less* than the benign comparator does, by about 0.17 nats per target. The result is stable to the convention and to the bootstrap. It is negative under (b), (b’), (c) and (d), and it agrees with a t -test ($p=0.013$) and a signed-rank test ($p=0.007$).

It would nonetheless be an error to report that as “the attack loses to budget-matched random paraphrasing”, and we do not. The comparator in that row is the benign candidate that maximises the *judge* score, chosen from 50; the attack’s judge score is an unselected re-scoring of a query chosen on NLI. The benign arm is handed a search the attack never ran, and that search is worth $+0.49$ nats

of selection premium in that arm. The NLI-selected benign candidate attains the judge maximum on only 37 of those 77 targets. Scoring both sides the same way, by taking the benign candidate that maximises NLI and reading *its* judge move, the net is $+0.016 [-0.112, +0.150]$ with a sign test at $p=0.66$; the exact-match arm moves the same way, from -0.027 to $+0.018$. Against the mean benign draw the same data give $+0.175 [+0.045, +0.312]$, and the attack’s judge score sits at the 0.70 mid-rank percentile of its target’s benign draws. The sign of the adjudicator’s “attack effect” is therefore a property of how the null is budgeted and selected, not of the attack: the interval clears zero above against a single draw, contains zero against a matched selection, and clears zero below against a judge-targeted best-of-50. We do not read the first of those as a win either. Its sign test does not reject ($p=0.44$), so its mean is carried by a minority of large movers. The null control exists to correct one defect, a maximum compared against a differently-budgeted maximum, and that defect reappears here one level down, per arm rather than per design. This time it runs *against* the attack rather than for it, which is very likely why review aimed at pro-attack bias did not surface it. **The defensible reading of the adjudicator arm is that its index-matched interval covers zero.** By the rule stated above we report it as covering zero rather than as null. An interval spanning $[-0.112, +0.150]$, about a quarter of a nat, does not license that stronger word. What it does support is narrower: at this n , on this population, and with the comparator matched, we cannot distinguish the optimised attack from a budget-matched benign search. Even index-matched the attack retains 181 NLI evaluations against the benign arm’s 50, so that reading is not a claim in the attack’s favour either.

Transfer and defense. These results are pending. Both the $SE \leftrightarrow SRE$ transfer table and the defense curve for reformulation-averaging fill from the same campaign. Both are reported net of the benign floor, and both read as corroborating rather than independent evidence, because the two detectors share the DeBERTa backbone.

5 Discussion

Hypothesised mechanism. Semantic entropy measures uncertainty over *meaning* by clustering sampled answers with an NLI equivalence relation. Its foundational assumption is that this meaning is a property of the question, invariant to how the question is phrased. Our attack targets exactly that assumption: a paraphrase q' that an NLI model still calls equivalent to q can nonetheless shift the target model’s answer-sampling distribution, and hence the induced cluster structure and its entropy. The false-alarm direction exploits the detector’s tendency to over-fragment a correct answer that can be expressed many ways; the hide direction exploits the mirror, collapsing a wrong answer’s diverse phrasings toward apparent consensus. In both cases we hypothesise that the manipulation lives in the gap between two notions of equivalence that the detector implicitly assumes coincide. One notion is the NLI gate’s and the other is the sampling distribution’s. Isolating that this gap is the cause rather than a correlate is exactly what the independent-clusterer check would establish.

How much of this is an attack, and how much is the detector? The mechanism above presumes there is room to move the score. At the top of the scale there often is not, and that changes what the false-alarm result is a result *about*. We should be precise about the scope, and about which population each number comes from: on the score-independent *fair* pool (200 correct, 200 hallucinating) the detector separates the two moderately well (AUROC 0.704 [0.653, 0.753]), so this is not a claim that discrete semantic entropy fails as a detector. It is a claim about the region of its range where a deployed threshold has to sit, and it is best stated in the unit an operator actually specifies: a false-alarm budget.

The achievable false-alarm rates are a short finite list. A detector that flags when the score reaches τ changes its false-positive rate only when τ crosses a value some clean correct answer actually took. So on any given population the rates it can be operated at form a finite set. That set holds one rate per distinct clean score, plus the always-available policy that never fires, and it is not spread evenly. Enumerated on the fair pool’s 200 correct answers, the achievable rates below one in

four false alarms are exactly three. They are 0%, 9.5% [6.2, 14.4] and 21.5%, catching respectively 0%, 27.5% [21.8, 34.1] and 44.5% [37.8, 51.4] of that pool’s 200 hallucinating answers. Counted against the budget an operator would state, the number of firing operating points at or below 1%, 5%, 10% and 20% false alarms is 0, 0, 1 and 1. We count against a budget rather than reporting how many operating points exist in total. More negatives realise more of the lattice, so the total is a property of the sample, while the low end is not. Sun et al. [27] argue that a score can rank acceptably and still leave an operator only a handful of usable thresholds. We instantiate that argument for a sampling-based detector, where the thresholds available are fixed by the sample budget rather than by a model’s verbal habits.

The lower end of that list is an *identity* rather than a measurement, for as long as the ceiling carries mass. The largest value the estimator can emit is $\ln N$, so every threshold above it flags nothing and, while some clean answer reaches it, the first threshold that fires at all is $\tau = \ln N$, which flags exactly the answers at the ceiling. The minimum non-zero false-positive rate is then not a tuning choice: it is $\Pr(\text{a clean correct answer yields } N \text{ mutually distinct meanings})$, the mass of the ceiling atom. At $N=10$, and under the DeBERTa-large-MNLI clustering that decides which meanings are distinct, an operator who specifies a 5% false-alarm budget cannot have one: the only threshold that honours it is the one that never fires, catching nothing. Both qualifiers are load-bearing. The budget-scaling paragraph below measures what lifting the sample budget costs, and the equivalence-relation paragraph below measures how far the clusterer alone moves the floor. Nor is the shortfall a property of our sample. The Wilson lower bound on the fair pool’s correct stratum is 6.2%, and the 1424-answer superset that stratum is a subset of estimates the same parameter with seven times the negatives, at 10.5% [9.0, 12.2].¹ A 10% budget we do *not* claim to exclude. It falls between the two populations’ estimates. It is honoured at 9.5% on the fair pool’s correct stratum, unavailable at all on the superset, and the intervals overlap either way. The defensible statement is therefore that the floor is about one correct answer in ten and a 10% budget sits on the boundary of feasibility. The 5% claim needs no such hedge on either population.

Prevalence changes the consequences of a chosen operating point but not the grid, which is a within-negatives object and so does not move when the mix of correct and hallucinating questions does. What does move is the rest of the arithmetic, so we state the operator’s experience on the full labelled pool throughout: its own floor of 10.5% [9.0, 12.2] on 1424 correct answers, its own true-positive rate of 25.2% [21.8, 28.9] on the 576 hallucinating ones at the threshold that floor sits at, and its natural hallucination rate of 28.8% (576 of 2000). The fair pool cannot supply this row. It is 200 correct against 200 hallucinating by construction, a mix the sampler chose rather than one the world presented, so it has no natural prevalence to state. Running at the lowest firing threshold then raises 148 alerts per thousand questions (295 of 2000), of which 49.2% [43.5, 54.8] are genuine (145/295), and leaves 216 hallucinations per thousand unflagged (431 of 2000).

What makes that gap a gap, rather than an interval a finer threshold could subdivide, is that the ceiling is an *atom* and not merely a maximum. At $\tau = \ln 10$ the flagged set is a single block of items carrying an identical score. That block is 19 of the fair pool’s 200 correct answers together with 55 of its 200 hallucinating ones, and no threshold can separate them, so those 19 false alarms are not a tuning failure but the fixed price of those 55 detections. The same holds one step down, where the second-largest attainable value moves the whole grid from 9.5% to 21.5% false alarms in one jump. This is what the coarseness of the lattice near the top of the range amounts to operationally: the two largest of the estimator’s 39 attainable values are not a small slice of the scale but two indivisible blocks of population, and an operator’s entire low-false-alarm menu is to take neither, one, or both. The entries of that menu are *points*, not a curve, and the finite achievable set they form grows with the sample.

The floor belongs to the whole detector, equivalence relation included. The floor identity above has a term in it that we have so far left implicit. “ N mutually distinct meanings” means distinct *under*

¹False-positive rate is a within-negatives quantity, so the 400-item fair pool has the *same* grid as its correct stratum and there is no separate pooled figure to report; what the hallucinating stratum supplies is the true-positive column. The superset is the full labelled pool’s correct stratum, of which the fair pool’s is a strict subset, so it pins the same parameter rather than a different one. That is what makes it a precision check on the floor and not a second population with a second answer.

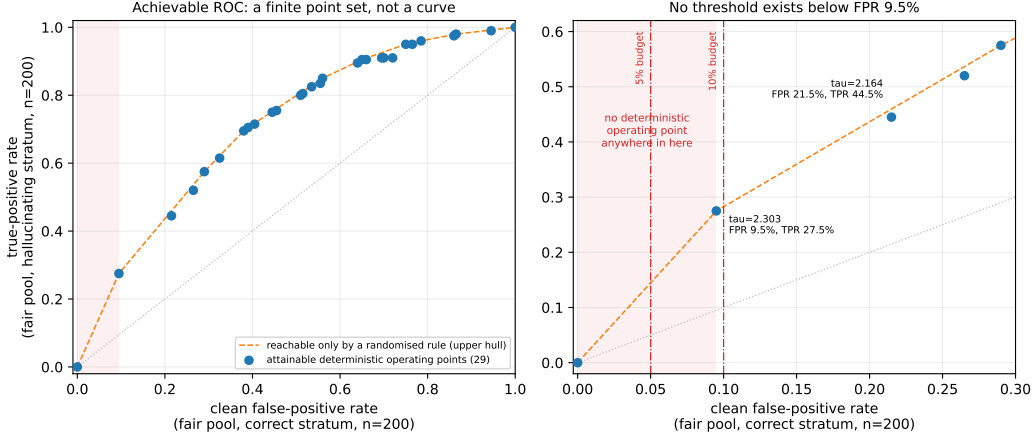


Figure 1: Every operating point a deterministic threshold on discrete semantic entropy can occupy at $N=10$, on the fair pool’s 200 correct and 200 hallucinating answers. Dashed: the upper convex hull. No deterministic threshold exists in the shaded region.

some equivalence relation, and the relation is a component of the detector rather than a property of the model, so the atom whose mass is the floor is a joint fact about the two. Our own campaign puts a number on how much that matters. Table 1 reports the same clean saturation rate three times over one fixed set of samples, the 80 correct-answer targets of the false-alarm stratum, with nothing varying between the three but the relation that decides which of a question’s answers count as the same meaning. Under exact match it is 46.2% [35.7, 57.1]; under the DeBERTa-large-MNLI clustering that defines the score it is 10.0% [5.2, 18.5]; under the LLM judge it is 1.2% [0.22, 6.75], which is a single target out of 80. The first two intervals do not overlap. The relation therefore moves this estimand by a factor of thirty-seven on identical samples, and it moves it across the very 5% line the identity above calls unreachable. So we scope every floor claim in this paper to the relation that produced it, and the scoped form is the one to carry forward: at $N=10$, under bidirectional DeBERTa-large-MNLI clustering, a 5% false-alarm budget is unavailable.

We do not read the converse off the same table, and the reason is the denominator. The judge’s rate is one target in 80, an interval reaching 6.75%, so these numbers do not establish that a 5% budget *is* available under a more permissive relation; they establish that the floor is not identified until the relation is named. Nor would a lower floor be free if one were confirmed. A relation that merges more readily lowers every score it touches, the hallucinating answers’ scores included, and every target measured here is a correct answer, so nothing in this campaign says what that trade does to detection. The point of the row is not that one of the three relations is right and the others wrong. It is that the floor is a property of the detector as a whole, its equivalence relation included, rather than of the language model alone, and that a floor reported without naming the relation has under-specified its own estimand.

What this does and does not say about the ROC curve. It does not say the ROC curve is fictional, and we state the limit of the claim before a reader has to. A *randomised* rule flags a ceiling item with probability p and otherwise never fires. Such a rule reaches any point on the chord between two adjacent achievable points, so a 5% false-alarm rate is attainable at $p=0.526$, for a true-positive rate of 14.5% against the 27.5% available at the fair pool’s own 9.5% floor. A randomised rule therefore buys sub-floor false-alarm rates only by paying the average of an indivisible block. Three things survive that concession. Every *deterministic* operating point, which is the only kind a deployment can be audited against and the only kind under which the same input reliably gets the same decision, is one of them (Figure 1). On the chord the operator pays the *average* of an indivisible block rather than selecting its most informative members, which is exactly what the granularity costs. And randomisation invents nothing below the chord’s left end: at any false-alarm rate the achievable

true-positive rate is capped by the upper convex hull of those points, and below the floor that hull is one straight line out of the origin.

Does a larger sample budget fix it? Half of this was settled without data and half needed the measurement, and the two halves are easy to conflate. Settled without data: the floor is non-increasing in N . Couple the budgets by taking the $N=10$ sample to be the first ten draws of the $N=20$ one; if all twenty answers are pairwise inequivalent then so are any ten of them, so {saturate at 20} is contained in {saturate at 10}, and the first ten of an i.i.d. draw of twenty have the law of an i.i.d. draw of ten. Raising the budget can therefore only lower the false-alarm floor, and the floor itself is a fact about how often the model answers a question in N ways the clusterer holds apart. That frequency is a property of the model, the question distribution and the equivalence relation jointly, not of the estimator’s arithmetic. Not settled that way: whether the grid becomes useably *fine* near the operating region. Enumeration bounds the number of thresholds available in the top tenth of the range at seven (455 attainable values at $N=20$, against 39 and two at $N=10$), but says nothing about the false-alarm rates those seven carry, and that depends on how the negatives distribute over them. Mass could spread evenly, giving a usable grid, or stay concentrated on the cap, giving the same pathology one budget along.

We have now run the budget out to $N=40$ over the fair pool, recording the full pairwise equivalence verdicts so that any smaller budget is recoverable by replaying them on random subsets, and the grid does become fine. Read off that one source, so that no step in the sequence is also a change of cache, the achievable false-alarm floor on the fair pool’s 200 correct answers falls from a replayed 12.0% [8.9, 15.3] at $N=10$ through a replayed 3.1% [1.8, 4.7] at $N=20$ to a measured 2.0% at $N=40$, a fall of 10.0 points [7.2, 12.9] that excludes zero. That last interval is a paired bootstrap over the questions and not the two end intervals subtracted, because the two floors are read off the same 200 questions and, at $N=40$, off a top score that can move to a different question when those questions are resampled. The ladder is monotone but it is not detectable at every rung, and we say so rather than let the sequence imply that it is. Taken leg by leg on the same paired bootstrap, the $N=10$ to $N=20$ leg is a fall of 8.9 points [6.8, 11.1], which excludes zero; the $N=20$ to $N=40$ leg is a fall of 1.1 points [−0.2, 2.4], which *covers* it. The end-to-end fall is real and its final increment is not separately resolved at 200 questions; both belong in the claim, and the second is the kind of thing this section elsewhere refuses to read a sign off. Figure 2 plots the three budgets and the three paired differences together. The first two entries are replays of the $N=40$ run’s recorded verdicts on random subsets, which is why they are quoted from that run rather than from the direct Week-4 cache the rest of this paper uses, where the same $N=10$ floor over the fair pool’s 200 correct answers is 9.5% [6.2, 14.4]; the two paragraphs below are about that difference. The figure shows that direct point separately, and joins it to nothing, because the June cache it comes from is a different generation run. Which interval a row takes is decided by the estimator and not by whether the row was replayed. Wilson prices a count of independent indicators at a threshold fixed *before* the data, so it belongs to the two rows that are counts of exactly that kind. Those two rows are the direct $N=10$ floor, 19 of 200 at $\ln 10$, and the empty at-cap mass at $\ln 40$. The replayed rows take a bootstrap over the 200 questions instead, because their point estimate is an average over subset draws rather than a count. Those resample a draw of questions and nothing else; the paragraph below is about why the replayed rows may not also resample the subset. The measured $N=40$ floor takes neither, for the reason below. What changes qualitatively at the top budget is that the floor stops being the ceiling atom: no clean correct answer reaches $\ln 40$ at all, so the at-cap mass is 0/200, or 0.0% [0.0, 1.9], and the floor becomes an ordinary order statistic. That statistic is the four highest-scoring clean answers, 2.0% of the fair pool’s correct stratum, at a threshold that catches 6.0% [3.5, 10.2] of that pool’s 200 hallucinating answers. That is also why we quote no interval for it. Its threshold is not $\ln 40$ or any other value fixed in advance but the top score *this* sample attained, and once the atom is empty nothing in the sample settles whether that score is the top of the population’s support. If it is not, a larger pool reaches a higher rung and reports a smaller floor, so the quantity being estimated moves with the pool instead of holding still to be estimated; if it is, the floor is an ordinary population proportion and the count in front of us estimates it. Nothing here decides which, and the two readings are far apart. Both candidate intervals inherit that indeterminacy, and each also has an endpoint placed

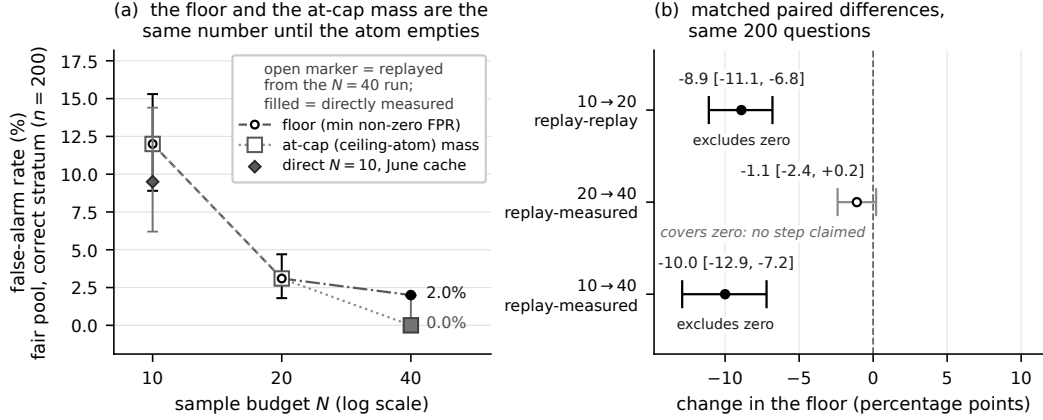


Figure 2: The achievable false-alarm floor on the fair pool’s 200 correct answers against the sample budget. Left: the floor, and in grey the ceiling-atom mass. Right: the three matched paired differences over the same 200 questions.

by construction rather than by the data. A question bootstrap over these questions cannot return less than $1/200$ however small the floor really is, and Wilson on a count of four cannot know that the rung it counts may not be the population’s top one. Neither prices the floor, so we report 2.0% as the cheapest alarm 200 answers can exhibit and leave it without an interval; what carries one is the at-cap mass above, whose threshold really was fixed in advance. An empty atom is not a free operating point, and we separate the two because the arithmetic invites the confusion: 0.0% is the never-fire policy, and the cheapest alarm that exists at $N=40$ costs 2.0%.

The grid also densifies where an operator would use it. At or below a 5% false-alarm budget $N=10$ offers no firing threshold at all, on either source, against five at $N=40$; at or below 10% the direct $N=10$ cache offers exactly one, the 9.5% point, against fourteen. So the 5% budget that $N=10$ cannot honour at all is honoured at $N=40$, at an achieved 5.0% [2.7, 9.0] on the fair pool’s 200 correct answers. The unqualified form of the claim above is therefore false and we scope it rather than defend it: the absence of any false-alarm rate between zero and roughly one correct answer in ten holds at $N=10$ and under this clusterer, and quadrupling the sample budget buys the missing operating point.

We report the achieved rate as a measured point and not as a certificate, and we read the pre-registered criterion on the quantity it was pre-registered on, which is not the one this paragraph used to compare it against. The plan required a Wilson upper bound below 5% on the count of correct answers *at the cap*, and derived from that a budget of at most three such answers in 200. It identified that count with the floor because at every budget it was written against, the two were the same number. At $N=40$ they came apart: the at-cap count is 0, not 4, its Wilson upper bound is 1.9%, and the criterion is met with three counts to spare. We disclose that reading rather than bury it, because the plan did not anticipate the atom emptying, and because comparing the four first-firing answers against a threshold derived for the at-cap count compares two different quantities. That is the same conflation this work has already had to correct once in the grid report. What the criterion licenses is narrow and we keep it narrow: the obstruction that closed off a 5% budget at $N=10$ is gone at $N=40$. That obstruction was an atom of mass at the ceiling, which no threshold can cut into. The criterion does not license reading the 5.0% achieved on the fair pool’s 200 correct answers as a demonstration that a sub-5% operating point exists, and we do not read it that way. One assumption holds the certificate up and we state it rather than leave it to be found: that some question in the population yields 39 or 40 mutually inequivalent answers out of 40 with positive probability. Were that probability exactly zero, the top of the support would be the rung this pool reached and the floor would indeed be 4 of the fair pool’s 200 correct answers; but 3.1% of them already yield 20 mutually inequivalent answers out of 20. The plan priced the remedy in advance too: thirty further correct answers along the same seed-0 ordering, taken from the larger correct stratum this one is a subset of, at the pessimistic end

of its own predicted floor, with the instruction to trigger the extension on the measured floor rather than on a hunch. The measured floor triggers it and we have not run it. So the concession is made at the strength its sample supports. That strength is five firing thresholds at or below 5% where $N=10$ offers none, and a floor of 2.0% that we do not certify to sit below 5%. The achieved rate carries the same caution from the other side, since the threshold that honours the budget is chosen on the same 200 correct answers whose false-alarm rate it is then quoted at.

We pre-registered where the claim would break. Before any of these samples were drawn we predicted, from the $N=10$ cache alone, that the ceiling-atom mass would fall to 3.5–5.3% at $N=20$ and 1.2–2.4% at $N=40$; we recorded in advance that the $N=20$ prediction sits on the very 5% boundary this claim turns on, and we committed to the reversal before the data existed, so that neither branch could be written after the fact. The plan’s own wording was “if it comes in far below, the finding becomes ‘here is the budget that buys a 5% false-alarm rate’”. The branch that fired is the one that was written down, but it fired only at $N=40$, and the prediction that led to it did badly. Both budgets came in *below* their bands, a replayed 3.1% at $N=20$ and an empty atom at $N=40$, and the plan had stated in advance, in bold, that its numbers were lower bounds on the atom and so erred if at all towards predicting too much relaxation. The pre-registered direction of the error is therefore itself falsified, which is worth more to a reader than two band misses are. At $N=20$ we claim nothing from the miss, and the reason is narrower than the row’s interval covering the prediction. The row is a replay of an estimator shown unbiased below, but it is still a replay, and its interval does *not* cover the whole predicted band: it reaches 4.7%, short of the band’s top at 5.3%, while containing the band’s lower edge of 3.5%. So the top of the prediction is excluded and its bottom is not, which is a partial miss and not the plan’s “far below”: a band whose lower edge survives has not been shown to be wrong, and a row we have just said may carry no load-bearing claim cannot be turned into a falsification either. At $N=40$ the atom emptied, a 5% operating point appeared, and the reversal clause is the one that applies. Even there the band is missed only on the quantity as it was predicted, since the plan identified the atom with the floor. Read as a floor, the measured 2.0% on the fair pool’s correct stratum lands inside the predicted 1.2–2.4%. The two parted company exactly when the atom emptied. We give the prediction not as decoration but because it is what separates a scoped claim from a retreat.

Two limits on this, and neither of them is a bias in the replay. The first is provenance. Only $N=40$ is directly measured; the smaller budgets are *replays* of that run’s recorded verdicts on random subsets, so we route the concession itself through the measured $N=40$ row and let no load-bearing claim rest on a replayed one. That is a rule about where a number came from, not a doubt about the estimator, which is exactly unbiased for the smaller budget: the 40 samples drawn for a question are i.i.d. and hence exchangeable, a uniformly random k -subset of an exchangeable 40-tuple has the law of k i.i.d. draws, and the clusterer is union-find over *pairwise* verdicts, so a subset’s clustering is exactly the clustering a direct k -sample run on those samples would have produced. What reusing one verdict matrix costs is the independence of the replicates and not the mean: every replicate conditions on the same 40 samples and the same 200 questions, so the spread across replicates is subset-draw noise around a conditional mean. That is a reason to distrust a sign test taken across replicates. It is not a reason to widen an interval, and the arithmetic of why is worth setting out, because the mistake is easy to make in the direction of apparent caution.

The floor a single replicate reports is a mean of independent indicators, one *per* question, question i saturating with probability p_i under the subset draw; so its variance splits as $p(1-p) + \text{var}(p) = \bar{p}(1-\bar{p})$, identically. The left term is the subset draw and the right term is the draw of questions, and their sum is exactly the binomial variance a Wilson interval on that one replicate already reports. On the fair pool’s 200 correct answers at $N=20$ the two terms, each written as the standard deviation it contributes to the floor and in that order, are 0.98 and 0.74 points against a binomial 1.23; at $N=10$ they are 1.61 and 1.63 against 2.30. Square-rooted like that they combine as $\sqrt{0.98^2 + 0.74^2}$ rather than by addition, which is the identity above with every term square-rooted and not a sum that fails to balance. The subset draw is therefore not missing from a single-replicate interval, and cannot be added to one.

Which interval belongs to which estimator follows. We quote the subset-averaged replay floor and never a single replicate. That floor is 12.0% at $N=10$ and 3.1% at $N=20$, each value the mean

of the 200 per-question p_i to a Monte-Carlo error under 0.01 points. The subset-averaged estimator has the subset draw averaged out of it, so its interval must not put the subset draw back: what is left varying is which 200 questions were drawn, each contributing an independent p_i , and a bootstrap over questions is the whole of it. The direct $N=10$ row is a count of independent indicators at the fixed threshold $\ln 10$ and takes Wilson; the measured $N=40$ floor is a count at no fixed threshold at all and takes neither, for the reason given above. The tempting error, and the one this arithmetic rules out, is to interval the subset-averaged floor by a bootstrap that resamples targets *and* redraws the subset: that counts the subset draw twice, once inside the target resample where it already sits and once again in the redraw, and on this data it returns intervals about twice the width the quoted estimator carries. Widening an interval is not the safe direction when the interval is the evidence, because it converts a measurable difference into a null. That is why we state, for every rate in this section, which estimator the interval belongs to. None of this bears on the unbiasedness argument above: it concerns the width of an interval and not the location of a point.

The second limit is a provenance question we have not closed. At $N=10$ the floor on the fair pool’s 200 correct answers is 9.5% [6.2, 14.4] measured directly on the Week-4 cache against 12.0% [8.9, 15.3] replayed from the $N=40$ run: a step of +2.5 points whose paired interval, $[-0.7, +5.5]$ over the questions, covers zero. The direct value sits near the fifth percentile of the replay distribution over 200 subset draws, which puts it low in that distribution and not outside it. Subsetting cannot produce the step, by the argument above, so two candidates remain. Sampling noise in the direct arm is one: that cache has 19 of its 200 correct answers at the $\ln 10$ cap, fewer than the replay predicts, at an exact one-sided Poisson-binomial $p=0.081$, with the rest of the cluster-count distribution a good fit ($\chi^2=7.39$ on 9 degrees of freedom, $p=0.60$). A real difference between the two generation runs is the other, and one is measurable: the August checkpoint’s answers run about 3.6 characters longer than the June cache’s ($z=+4.0$) and end in terminal punctuation 1.6 points less often. What we cannot do is read that drift against a configuration held fixed, because the two runs’ artefacts record only the token budget and the seed in common: past the model id, its 4-bit loading, temperature and that budget, the June manifest pins little of what decides the model’s output. It does not pin the nucleus, the quantisation type or the model revision. The August checkpoint records nothing of its generation settings at all beyond the sample count, the token budget and the seed (Section 6). The drift may be run-to-run variation under configurations that did match, or a configuration difference we cannot see, and nothing in either artefact separates the two. Longer and more often truncated answers entail each other less often, which pushes the ceiling atom up, so the sign is the one the gap would need; how large that effect is we have not measured, and the cluster-count distribution’s failure to separate the two runs does not bound it (Section 6). We therefore cannot separate subsetting error from generation drift, and we claim no direction for the gap. One clean direct $N=10$ pass on the current machine separates them by subtraction. It is one of two measurements this section is short of, both of them cells the budget plan pre-registered in advance and neither of them run: this one, a fresh $N=10$ pass over 40 answers of the correct stratum, and a direct $N=20$ pass over 60 answers of the same stratum. The second pass asks the other half of this question. Does a replayed grid match a directly measured one at the same budget?

The one $N=20$ re-score elsewhere in this work stands in for none of the above. On 15 saturated targets re-scored at $N=20$, lifting the cap by 0.693 nats left 20% of them still pinned and returned a mean of +0.304 nats of usable headroom rather than the full +0.693. We report that shortfall without a mechanism for it. The obvious candidate is that the clean baseline climbs along with the cap, and it is not ours to offer. A plug-in entropy that rises towards its estimand as the sample grows is that estimator’s negative bias decaying, which is the coverage-adjustment account this paper cites in *support* [17, 23], and one finding cannot be both our explanation and their result. The pilot is a measurement of *headroom*, not of the grid: its 15 targets were chosen because their *attacked* $N=10$ score hit the cap, which is selection on a score and on the attacked side, and at that size the interval around the ceiling rate spans almost everything of interest. It is not evidence about the grid in either direction, and nothing else in this work is either: no direct measurement of the grid at $N=20$ exists. On the fair pool’s 200 correct answers the trend above reports 3.1% at that budget, and that entry is a replay of the $N=40$ run’s recorded verdicts on random 20-subsets; the only answers this project has ever scored at $N=20$ are the 15 just described, and they were selected on a score. The cell that

would settle it is the direct $N=20$ pass named in the paragraph above, pre-registered for exactly this purpose and not run.

Under the attack’s own success criterion the same coarseness bites from the other side. A target whose clean score already sits at one of the top two attainable values cannot register a success whatever the paraphrase: the two lie 0.139 nats apart, so the criterion’s $\delta=0.25$ is out of reach from the lower one and there is nowhere above the upper one to go, which disqualifies 21 of the 80 correct-answer targets of the attack campaign before any paraphrase is generated (Methods).

The sharpest way to put the finding is therefore not “we constructed an attack that defeats semantic entropy” but “at the standard $N=10$, and under the DeBERTa-large-MNLI clustering that defines it, discrete semantic entropy offers an operator no false-alarm rate between zero and roughly one correct answer in ten, and the false-alarm direction is correspondingly unmeasurable in the detector’s own units.” The two halves are the same fact seen from the two ends of the deployment, and the first is the stronger form because it is a statement about what does not exist rather than a frequency we observed. Every qualifier in it is load-bearing and we drop none of them. *Discrete* carries one: the ceiling behind this reading is shared with the likelihood-weighted estimator, but the lattice that makes the range so coarse is not, and the discrete grid of achievable false-alarm rates that comes with it is not shared either (Section 3). The first claim invites the reply that a stronger detector, or a stronger optimiser, would settle the matter. The second does not, because the natural rebuttal is to use an estimator with more range, and that rebuttal concedes the point. The rebuttal is already being built: coverage-adjusted estimators exist precisely because the plug-in statistic misrepresents the quantity it estimates at deployed N [17, 23]. Whether the remaining, uncensored margin is one an adversary can reliably exploit is the question the confirmatory evaluation answers. It has now answered it, and we report the answer with the prominence we promised it whichever way it fell: on the false-alarm targets the pre-registered exceedance test does not reject in any clusterer arm (Section 4). No exploitable margin is demonstrated. That is a statement about what we can show at this design’s power, not a demonstration that none exists, and the two must not be run together.

That reading also disciplines what we may say about magnitude. Because roughly half of attacked targets end at the cap, the attack and any other method that also saturates become indistinguishable there, and effect sizes expressed in nats are censored. The cap bites asymmetrically. On the fair pool 27.5% of hallucinating answers sit there against 9.5% of correct ones, which is the same pair of numbers as the lowest firing operating point above, read from the other direction. We are careful not to over-read that. AUROC is a rank statistic and $\min(\cdot, \log N)$ is monotone, so censoring changes no pairwise ordering except among pairs where *both* members are pinned. That is $0.275 \times 0.095 = 2.6\%$ of pairs, bounding the downward bias at 0.013 against a confidence half-width of 0.050. The asymmetry in fact makes the effect *smaller*: had both strata been censored at the higher of the two rates, $2.9\times$ as many pairs would be affected. The consequence of the asymmetry is therefore about which class loses resolution, not about the reported AUROC. And because the reported effect is the maximum over a large candidate search, a further part of it is selection on estimator noise, and on our own data that part is measurably about half. Strip both away and what remains is a modest and heavily conditioned quantity. We think the honest summary is that whatever margin exists is small, and that most of the apparent margin in an uncorrected report is an artefact of how the number was obtained. Whether the remainder is real is the question the confirmatory run decides. It has now run, and it does not decide it in the attack’s favour: the pre-registered statistic does not reject, so, consistent with Methods, we claim no attack effect here.

Is the weakness structural? We conjecture the mechanism generalises beyond one implementation: any detector that operationalises “semantic equivalence” through a learned model inherits that model’s decision boundary as an attack surface. The very component that defines the score is the component an attacker steers. This *a-priori* argument follows from the construction alone; the load-bearing *empirical* test is whether the effect survives an equivalence oracle independent of the detector’s own NLI. We run that test. Cheap independent oracles do not settle it. Exact-match saturates on short factoid answers, and sentence embedders are near-chance on the adversarial, word-preserving case (Methods). A victim- and NLI-independent LLM-judge, validated to 0.93 accuracy on the gold-alias hard negatives that proxy that adversarial discrimination (where the embedder scores

0.51), can settle it for the false-alarm direction, under the disclosed re-scoped gate (Methods). Under it, the structural question now has an answer for one direction of one detector, and the answer is negative. An earlier $n=6$ machinery-validation sample is evidence about the pipeline rather than about the mechanism: its interval includes zero, and its per-target diagnostics are too coarse at that sample size to order the clusterer arms (Methods), so we quote no point estimate from it and draw no reading either way. The confirmatory run has since completed on all 80 false-alarm targets, and the adjudicator’s pre-registered exceedance test does not reject (Section 4): with the attacker’s own search budget priced into the null, the optimised search does not beat a budget-matched benign one under an equivalence oracle the detector does not supply. We report that as the gate said we would, and we decline the stronger form of it. A non-rejection by a design with power 0.77 against a two-fold effect bounds what we could have seen, not what exists; it is confined to the false-alarm direction of the SE detector on those 80 targets, and it leaves the hide direction and SRE untested. What the run removes is the licence to claim the effect, not the possibility of one. A single validated judge, however independent, is in any case not the last word; SE-to-SRE transfer, confounded by the shared DeBERTa backbone, is at best corroborating.

The null control’s own result is a measurement-validity result. The adjudicator arm does not report one number, it reports a family, and the family does not agree on a sign. Against the mean benign draw the attack’s paired net under the LLM judge is $+0.175$ [$+0.045$, $+0.312$] nats; against the benign candidate selected the same way the attack’s own query was selected it is $+0.016$ [-0.112 , $+0.150$]; against the best of 50 benign candidates ranked on the judge itself it is -0.173 [-0.302 , -0.039]. The attack, the targets and the oracle are identical across those three rows. The only thing that changes is what the attack is being compared against, and it moves the interval from excluding zero above to excluding zero below. Not even the top row is a clean win, and we say so rather than borrow its sign: its bootstrap interval clears zero but its sign test does not ($p=0.44$), because the mean there is carried by a minority of large movers. That the same data can be read three ways is the point. This is the paper’s thesis arriving inside the paper’s own experiment: an attack effect quoted against an unbudgeted comparator is in large part a statement about the comparator, and the same is true of an attack *deficit* quoted against an over-budgeted one.

We hold ourselves to that in both directions, which is the part worth saying out loud. The -0.173 is the number a reader most wants us to keep, because it is the number least flattering to our own attack, and we do not keep it as a finding: the comparator that produces it is handed a judge-targeted search over 50 candidates while the attack’s judge score is an unselected re-scoring of a query chosen on the detector’s NLI. That is a maximum compared against a differently-budgeted maximum. It is the exact defect the null control was built to remove, reappearing one level down, per clusterer arm rather than per design. It survived because it ran the wrong way: a review looking for bias toward the attack has no reason to stop on a number that hurts the attack. A validity check with a direction is not a validity check, and we would not have caught this one by tightening the checklist we already had. Index-matched, the adjudicator arm cannot be distinguished from zero at $n=77$. That is what we claim. We do not claim the reversal, and we do not claim a win.

Why lead with false alarms. Red-teaming of *sampling-based* uncertainty estimators has emphasised the hide direction (missed hallucinations), though the inflation direction is well established against other confidence channels [10, 26]. We foreground false alarms because, for a deployed abstention or flagging system, systematically inflating the uncertainty of *correct* answers is its own failure mode: it erodes utility and user trust, and it is cheaper for an adversary who merely wants to make a model look unreliable. That this *class* of detector is exposed to it is what would have been new here, and the confirmatory run does not establish it: the exceedance test does not reject in any arm, so what we contribute in this direction is the protocol that makes the question answerable and a bound on the effect at this n , not the exposure. The direction itself was not new either.

Defence. Averaging the detector score over several reformulations of whatever input is received should dilute a single adversarial phrasing, and that averaging is the mechanism SRE already builds in.

We state that expectation as a prediction the experiment can refute. Because the attacked paraphrase was selected as a maximum over the feasible candidates of a ~ 181 -evaluation search, merely re-scoring it on fresh samples shrinks the effect by roughly half with no defence of any kind, so the defended arm has to be measured against a *cost-matched* control that pays exactly that re-sampling cost and nothing else. That control is a $d=1$ fresh redraw of the same two inputs. The claim under test is that the per-question paired reduction of the d -averaged arm against that control is positive with a 95% confidence interval excluding zero. Its failure condition is explicit: if the interval covers zero, reformulation-averaging has bought the attacker’s query cost rather than robustness, and the defence claim is false. That experiment has not been run. Its arms and its statistic are fixed in advance, and we report the verdict whichever way it falls. Everything above is to be read against a null noise floor, so that what we ultimately claim are effects exceeding what benign rephrasing achieves by chance. That floor has now been measured for the false-alarm direction and is reported in Section 4; the defence curve and the $SE \leftrightarrow SRE$ transfer are not covered by it, have not been run, and remain readings of preliminary data rather than claims.

6 Limitations

Our results are scoped to the discrete estimator, and one of them may not transfer. We study discrete semantic entropy, for the three reasons given in Section 3, and the paper’s claims do not all have the same standing under the likelihood-weighted estimator that Farquhar et al. [6] designate as their main method. The $\log N$ ceiling transfers by construction: their Eq. (5) normalises cluster likelihoods into a categorical distribution over the same at most N observed clusters, so its plug-in entropy lies in $[0, \log N]$ exactly as ours does. The 39-point lattice does not: real-valued weights are not confined to the entropies of integer partitions of N , so the exact ties at the cap, the 0.139 nats separating the top two attainable values, the unwinnable share of the false-alarm denominator, and the hard-tie machinery the exceedance test needs all lapse. Our headline result lapses with them, and we say so plainly. The finite grid of achievable false-alarm rates, and the floor beneath it, exist because the maximum $\log N$ is an *atom*. Under real-valued weights it is not one, because Eq. (5) attains its maximum only when the normalised cluster likelihoods are exactly equal. Whether a pile-up *near* the cap would leave a near-atom, and so a near-floor, is a separate empirical question, and the only part of this we can speak to without the real weights. Our own cached cluster structure covers the 1424 greedy-correct questions (alias-aware span oracle, as everywhere else in this paper) of our 2000-question replication pass, and with them the 200-question correct stratum of the fair pool nested within that set. Re-weighting that structure with simulated length-normalised likelihoods $\ell \sim \mathcal{N}(0, s^2)$ in place of sample counts separates two things that “the pile-up” runs together (`scripts/likelihood_weight_sensitivity.py`). The *exact* atom does not survive: at any positive spread no question sits exactly at $\log N$, and the achievable floor collapses to $1/n$. That is the measure-zero argument above restated as a measurement rather than a bracket, and no choice of s rescues it. Near-cap *crowding* does survive at small spread and decays smoothly: on the fair pool the share of clean correct answers within 0.05 nats of the top score is 9.5% at $s=0$, still 9.5% at $s=0.1$, and halves by $s \approx 0.46$. What we cannot say is where the real spread falls. Our cache stores sample strings and not likelihoods, so nothing we have measured constrains s for length-normalised sequence log-likelihoods from our quantised model on these answers. We therefore report the curve and decline to claim that its crossover falls inside a plausible range. Settling it needed the real weights, so we re-scored our stored samples under the generating model’s own sequence likelihoods, on the same questions and the same clusterings, and computed all three estimators side by side. The simulation’s prediction holds and sharpens. That comparison runs over all 2000 questions of the replication pass at natural prevalence, which is the 1424-answer clean-correct superset above together with its complement, and not the fair pool. With length-normalised likelihoods the at-cap atom is *gone*, at 0 of those 2000, 0.0% [0.0, 0.2], against 295 of 2000, 14.8% [13.3, 16.4], under the discrete estimator. Near-cap crowding partially survives, at 399 of 2000, 20.0% [18.3, 21.8] in the top tenth of the range against 556 of 2000, 27.8% [25.9, 29.8]. The atom, and therefore the false-alarm floor it induces, is a property of sample-proportion weighting; the crowding is not. Length normalisation

is the choice doing that work: on raw sequence likelihoods the cluster softmax is nearly one-hot and rank agreement with the discrete estimator collapses to $\rho=0.38$, whereas length-normalised it is $\rho=0.99$, and on the score-independent fair pool (200 correct, 200 hallucinating) it scores AUROC 0.703 against the discrete estimator’s 0.704. Every AUROC in this paper therefore stands under Eq. (5) as we compute it, and only the saturation arithmetic is estimator-specific.

One cached generation run underlies every $N=10$ number here. The re-scoring above, the achievable false-alarm floor and its grid, the clean AUROC on the fair pool, and the corresponding floor on the 1424-answer superset are all computed from a single cached pass of $N=10$ samples over 2000 TriviaQA validation questions, generated in June 2026. Two properties of that cache bound what any of them establishes, and neither is visible in the numbers themselves.

First, the run is under-recorded. Its manifest pins the model id, 4-bit loading, the sample count, temperature 1.0, a 48-token generation budget, the seed and the split, and of the settings that determine what the model emits it pins nothing further: not `top_p`, not the 4-bit quantisation type, not whether double quantisation was applied, not the compute dtype, not a model revision, and not the `transformers` version. Those six values are recovered from the defaults of the sampling script’s own source rather than read off the artefact. This paper quotes none of them as a property of the June run, and a later draft should not start. Exactly one place in this project has a value for any of them on record, the `top_p=1.0` in the $N=40$ run’s own provenance block. That block describes the August generation pass and says nothing about this cache. The absent revision hash is the one we would most like back: without it we cannot certify that a re-run today would load the same weights. A further gap is structural. The cache stores decoded answer strings and never the token ids the sampler emitted, so the likelihood re-scoring teacher-forces each answer under its *canonical* tokenisation; 17 of the 20,000 cached samples (0.085%) retokenise past the 48-token budget, which proves the two id sequences differ for at least those, and the detokenise-then-retokenise string round trip itself is exact for only 19,996 of the 20,000, its four failures all trailing whitespace. Both figures are lower bounds, since a resegmentation that preserves the token count leaves no trace in either check, so what Eq. (5) recovers above is the likelihood of each answer *string* under its canonical tokenisation rather than of the sequence the model emitted.

Second, generation runs of this pipeline are not interchangeable, and we have measured that they are not. The August $N=40$ run, drawing fresh samples for the same fair-pool questions, produces answers about 3.6 characters longer ($z=+4.0$) that end in terminal punctuation about 1.6 points less often; the two runs’ artefacts record only the token budget and the seed in common, so we cannot certify that the rest of their configurations matched either. We disclose that drift in Section 5, where it bears on a comparison between the two runs. It bears on every June-cache number in this paper as well, and we have not bounded it for any of them. The statistic that would carry it into them is the distribution of the cluster count at $N=10$, which is the whole of what the discrete score is a function of, and on the fair pool’s 200 correct answers that distribution is not distinguishable from what the August run’s recorded verdicts predict for it. We decline to read that as a bound: a comparison on 200 questions that fails to separate two runs is not a measurement of how close they are, and treating it as one is the move this paper has had to retract elsewhere. One fresh direct $N=10$ pass on the current machine separates drift from sampling noise by subtraction, and we regard it as owed.

Selection on a noisy estimator, measured rather than merely acknowledged. Semantic entropy is a finite-sample quantity: for each candidate we cluster $N=10$ sampled answers, so the score is a Monte-Carlo estimate of the true entropy. Our optimiser selects the extremum of this estimate over the feasible candidates of a ~ 180 -candidate search, and the extremum of many noisy-but-unbiased estimates is biased in the attacker’s favour even when, as here, sampling is seeded so each estimate is *reproducible*. Reproducibility is not the same as noise-freeness.

We quantify the resulting inflation directly, by re-scoring each *selected* paraphrase on an independent sample at the same N under a different sampling seed. That is the classical diagnostic for selection on noise, and it requires neither an oracle nor a benign floor. The re-scored set is the 69 of the 80 false-alarm targets on which the optimiser found a paraphrase at all; on the other 11 the search returned the original question, so there was no selection to re-test. Across those, the mean

intended move falls from $+0.609$ nats at selection to $+0.268$ nats on fresh samples. Two statements follow, with very different strengths, and we separate them. *That* the selection-time figure is inflated is demonstrated: the shrinkage is -0.341 nats with a bootstrap interval of $[-0.469, -0.212]$ that excludes zero. *How much* it is inflated is not tightly determined: retention is a ratio of means, and its interval is wide at 44% [23%, 64%]. So roughly half of the effect a naive report would claim is selection on sampling noise rather than a property of the paraphrase, but the honest range runs from under a quarter to two-thirds surviving. Signal remains either way, since 37 of the 69 keep a positive move and the two measurements correlate at $r=0.48$.

We therefore do not report raw effect sizes as the headline. The claim statistic prices the attacker’s search budget into the null (Section 4), so an effect that does not exceed what a benign search of comparable size achieves by chance is not claimed. We note that this re-scoring diagnostic bounds only the *selection* component: it says nothing about whether random rephrasing achieves the same move, which is the separate question the null control answers.

The clusterer and the equivalence gate share one NLI model. Semantic entropy is *defined* by DeBERTa-large-MNLI bidirectional clustering, and our attack’s feasibility constraint uses the same model, so a measured degradation could in part reflect that one model’s self-inconsistency rather than a paradigm-level weakness. We address this directly by re-clustering the answer samples under equivalence oracles independent of the DeBERTa-MNLI supervision (Methods), but two caveats bound the check. First, the cheap independent oracles are unreliable on the adversarial case (exact-match saturates; the sentence embedder is near-chance), so the adjudication rests on a single LLM-judge. That judge is validated (0.93) on the hard discrimination the others fail, but a single oracle is not dispositive even when it is independent and validated, and a human or multi-judge equivalence audit remains owed. Second, the judge’s own positive-recognition is imperfect, at 0.65 [0.59, 0.70]. On the clean gold-alias pairs it was validated on, it *over*-splits genuine aliases, in part correctly rejecting noisy TriviaQA alias labels. The pre-registered conservativeness argument for the false-alarm direction was built on that behaviour. Over-splitting inflates the clean baseline, so a surviving false-alarm effect is understated. **That argument does not survive contact with this campaign’s own data.** On the messy sampled answers the deployed judge actually clusters, it *under*-splits: over the 80 false-alarm targets its mean clean baseline entropy is 0.648 nats against the NLI clusterer’s 1.477, and it collapses all ten samples into a single cluster on 27 of the 80 clean baselines where NLI does so on 4. Both statements are true, each of a different input. The judge over-splits clean alias pairs, and it under-splits sampled answers. The second behaviour is the one that governs the numbers we report, so we do not carry the first forward as though it settled the direction of the bias. The consequence runs the other way from the pre-registered worry: a floored baseline bounds the measured false-alarm move below by zero on those 27 targets, which is anti-conservative for a false-alarm claim rather than conservative. We disclose it rather than correct it, because we did not make a false-alarm claim: dropping the floored targets leaves the judge-arm move at $+0.247$ [$+0.072, +0.429$] nats against $+0.291$ [$+0.152, +0.440$] on all 80, and the adjudicator’s exceedance test does not reject either way. The independent-clusterer result is reported as the confirmatory $n \geq 80$ run’s own result and no longer as preliminary evidence; what remains preliminary is the merge-side bias estimate, which the planned validation was designed to be blind to and which is owed before any judge-arm effect size is claimed.

One consequence of that shared dependence reaches past the attack and into the measurement-validity result, and we had not scoped it far enough. The achievable false-alarm floor we report is the mass of an atom at $\log N$, and what sits at $\log N$ is whatever the clusterer declines to merge, so the floor is a property of this clusterer as much as of the model. Section 5 measures how far it moves when the relation is changed and the samples are held fixed: on one set of 80 clean targets the same estimand reads three different ways, and the exact-match and NLI intervals do not overlap. Every floor claim in this paper is therefore scoped to bidirectional DeBERTa-large-MNLI clustering. We do not claim the converse, that a more permissive relation delivers a usable low false-alarm rate, because the one relation that would suggest it is measured at a denominator too small to exclude the budget in question.

Equivalence is certified automatically, not by humans. Feasibility is a judgement of bidirectional entailment. Our answer-flip subcategory collects the successful attacks in which the model’s correctness flips under q' , and that subcategory is a *lower* bound on how often the gate admits a non-equivalent paraphrase, since a q' can shift scope or presupposition while leaving the greedy answer unchanged. We do not yet report a human (or disclosed strong-LLM-judge) equivalence audit of a random sample of successful q' ; such an audit, with an inter-annotator agreement estimate, is needed before the meaning-preserving claim is fully established.

Status check versus what the detector sees. The success criterion re-checks the model’s answer with a single greedy decode, while the detector clusters temperature-1.0 samples. The two can disagree, because a greedy answer may stay wrong while the sampled distribution shifts toward correct (or vice versa). We therefore report both the greedy status and the fraction of the detector’s own samples that are correct under q' , rather than relying on the greedy check alone.

Model, precision, data, and budget. Results are on a single model family (Llama-3.1-8B-Instruct) run in 4-bit quantisation on one 16 GB GPU; quantisation may shift both the entropy estimates and the attack surface relative to a full-precision detector, and generalisation across model families and scales is untested. Our only dataset is TriviaQA, and no result here covers a second corpus. The attack budget (beam and iteration counts) is modest by necessity, and the reported cells use small per-stratum n , so the confidence intervals are wide and the numbers are best read as preliminary effect sizes rather than tight estimates.

One measured sample budget, and how coarse the ladder is. The budget in the paragraph above is the *attacker’s* search budget; this one is the detector’s sample budget N , and the scope limit we draw from it needs stating in its own right. The achievable false-alarm floor falls as N rises, so that the absence of any low false-alarm operating point is a fact about $N=10$ and not about semantic entropy. That claim rests on a single directly measured budget. We ran $N=40$ once, over the fair pool’s 400 targets, 200 per stratum. The $N=10$ and $N=20$ rows of the falling sequence are not further experiments: they are replays of that one run’s recorded pairwise equivalence verdicts on random subsets of its 40 samples, unbiased for the smaller budget by the exchangeability argument of Section 5, but read off one generation run rather than three. Only the top of the ladder is a measurement and nothing above it was run, so what we can say is that the floor is non-increasing in N , which we prove rather than measure, and that on the fair pool’s 200 correct answers it had reached a measured 2.0% by $N=40$; how it behaves further out, and whether the decay between two coupled points is the shape it keeps, we cannot say. Three narrower scopes ride along. The ladder exists only on the fair pool. The 1424-answer superset estimates the same $N=10$ floor as the fair pool’s 9.5% over seven times as many negatives, but it has no $N=40$ counterpart, so every row here is estimated on 200 negatives, a resolution that cannot separate false-alarm rates less than half a point apart. It is one model family in one quantisation on one dataset read through one equivalence relation, and the floor is a fact about how often that model answers a question in N ways the clusterer holds apart, a property of the model, the question distribution and that relation jointly rather than of the estimator’s arithmetic, which is exactly the kind of quantity that need not transfer to another model, another corpus or another notion of sameness. And what the extra operating points are *worth* we do not report in either direction: at a matched false-alarm rate the cross-budget change in true positives has a magnitude that depends on which decision rule is paired and a sign that depends on which $N=10$ cache is paired, and no pairing separates the budgets on 200 questions, so we claim no discrimination benefit from a larger budget and, equally, none against one.

SRE fidelity and threat model. We reproduce SRE with exact-match-then-NLI-union-find clustering rather than the paper’s full energy-based hybrid, and our NLI backend (deberta-large-mnli) is weaker than the deberta-v2-xlarge used in some prior work; the SRE target (paper 0.871) depends further on the clustering backend we simplified. Our *SE* replication reaches AUROC 0.694 under the greedy alias-aware span oracle this paper operates with, against the paper’s *SE* figure of 0.828. We

do not attribute that 0.134 gap to the backend. The nearer-looking 0.790 is the same run scored under an all-samples-correct label, a convention mechanically coupled to semantic entropy that we use nowhere else (Section 4); backend and correctness convention are thus confounded in the gap, and separating them needs an ablation we have not run. Because SRE re-paraphrases its input internally, the attacker’s chosen q' is only one input to a detector that then generates its own reformulations; the SRE threat model (whether the attacker controls the reformulation set) needs to be stated and matched to the implementation, which we flag as a limitation of the SRE results specifically.

Our novelty claims carry a documented search gap. Every “to our knowledge” in this paper is bounded by a coverage limit we can state precisely. We searched the public arXiv record, ICLR 2026, TMLR under-review, six ARR cycles, and the COLM and ICML workshop tracks. We could not search the NeurIPS 2026 main track, the ARR February, June and July 2026 cycles, the COLM 2026 main track, or the full text of any OpenReview submission. For those three ARR cycles the indices return zero results to control queries that return hits for neighbouring cycles, so we have verified the index is empty rather than merely finding nothing. Absence of hits from those venues is therefore *uninformative*, not weakly reassuring, and we do not treat it as evidence. The relevant pace: the closest concurrent work we identified had been public for six days when we found it [10].

7 Conclusion

Sampling-based hallucination detectors treat semantic uncertainty as a property of the question, invariant to surface form. We set out to test that assumption adversarially and found that the more informative obstacle lay upstream of the adversary, and that it is sharpest stated as a constraint on the operator rather than on the attacker. At $N=10$ the *discrete* estimator we study is confined to a lattice of 39 attainable values with an atom at the maximum, so a detector that flags when the score reaches a threshold can be run only at a finite list of false-alarm rates. Scored clean on the fair pool’s 200 correct answers, the achievable rates below one in four are 0%, 9.5% [6.2, 14.4] and 21.5%, catching 0%, 27.5% and 44.5% of that pool’s 200 hallucinating answers. An operator who specifies a 5% false-alarm budget cannot have one at that sample budget, because the only threshold that honours it never fires. That floor is an identity rather than a measurement for as long as the cap carries mass, since every threshold above $\ln N$ flags nothing and the smallest rate a firing threshold can have is therefore the mass of the ceiling atom itself. That mass is the probability that a clean correct answer yields N mutually distinct meanings, which the 1424-answer superset puts at 10.5% [9.0, 12.2]. At $N=10$ a 5% budget is excluded at 95% confidence on both populations; a 10% budget is on the boundary and we claim no more than that. The qualifier is load-bearing: measured out to $N=40$ over the same 200 correct answers the cap carries no mass (0.0% [0.0, 1.9]), the floor ceases to be an identity and becomes an ordinary order statistic at 2.0%, and the threshold that honours a 5% budget achieves 5.0% [2.7, 9.0] on that same correct stratum of the fair pool. We report the second as a measured operating point and not as an existence claim, and we attach no interval to the first: its threshold is the top score *these* 200 answers happened to reach, so 2.0% is the cheapest alarm a pool of this size can exhibit rather than an estimate of how cheap an alarm can be. What the pre-registered criterion asks for is a one-sided bound on the mass *at the cap*, and that bound is the 1.9% above, which falls under the line rather than over it (Section 5). A second qualifier is load-bearing in the same way, and it names the clusterer rather than the budget. “Mutually distinct” is whatever the equivalence relation says it is, and on the same attacked targets and the same clean samples the mass at the cap moves by more than an order of magnitude as that relation is varied from exact match to the detector’s NLI clusterer to the independent judge, the exact-match and NLI intervals not overlapping (Section 4). The floor we report is accordingly the floor under the bidirectional DeBERTa-large-MNLI clustering the detector ships, and what we claim across relations is that dependence itself rather than a floor surviving all of them. The claim is about deterministic rules: a randomised rule reaches the chords between these points, but only by paying an indivisible block’s average, and below the floor the reachable frontier is a single chord out of the origin. The $\log N$ ceiling behind all of this is shared with the likelihood-weighted estimator its authors designate

as their main method. The lattice, and so the grid of achievable false-alarm rates, is particular to the discrete variant that the same authors run for their GPT-4 results and that semantic entropy probes are trained to predict. This is not a claim that the detector fails. On the score-independent *fair* pool (200 correct, 200 hallucinating) it separates correct from hallucinating answers moderately well, at AUROC 0.704 [0.653, 0.753]. That fair pool is the wider population on which we characterise the detector, and the one our targets are a sub-sample of. The claim is instead that at $N=10$ the region where a false alarm lives is compressed to the point where an operator has no usable choice inside it, and differences there are largely unmeasurable in the detector’s own units.

Establishing even that much honestly turned out to require more care than the question first appears to need. Because the attack reports a maximum over a large candidate search, about half of its apparent effect is selection on estimator noise, which we measure directly by re-scoring the selected paraphrase on independent samples. Because the score saturates, effect sizes in its native units are censored on just over half of attacked targets, and a null must remain valid when attack and benign candidates tie at the cap. And because the detector’s clusterer is the same model that certifies our paraphrases as meaning-preserving, no single-clusterer measurement can separate a real change in the model’s answers from that model’s own inconsistency. The confirmatory evaluation was run under a protocol built to survive all three. Targets were chosen without reference to the detector’s own scores, success required the hallucination status to survive the paraphrase, and the null prices the attacker’s search budget in rather than comparing a maximum against single draws. We confine claims to what that protocol supports, and on the evidence it returned that is no attack effect. The pre-committed exceedance test fails to reject in every clustering arm. We draw that verdict from the pre-registered adjudicator arm rather than from the shared-NLI one, and the reason is a limit of our own instrumentation rather than a preference: the search budget the null prices in is the number of candidates the attack’s equivalence gate admitted, the gate was applied lazily, and the run recorded enough to bound that number from above but not to identify it. The shared-NLI arm’s p -value is sensitive to where in that range the true budget lies. The adjudicator’s is not, and it does not reject anywhere in it.

Two boundaries remain, and we state them as the work to do rather than as caveats to wave away. First, our clusterer and our equivalence gate are the same NLI model, so whether the vulnerability is a property of the *paradigm* or of one model’s self-inconsistency turns on an independent clustering oracle. We build one. The cheap candidates (exact match, sentence embeddings) fail on the adversarial case, but a victim- and NLI-independent LLM-judge, validated on a domain-matched proxy of that case (0.93 vs. 0.51), clears it. It is under this judge that the confirmatory large- n run, now complete, adjudicated whether the false-alarm effect survives the independent clusterer (the direction to which the judge is re-scoped; Methods). It does not survive it: the adjudicator’s exceedance test does not reject, and index-matched against a benign candidate selected the same way the attack’s own query was, the adjudicator arm cannot be distinguished from zero (+0.016 [−0.112, +0.150] nats). That is the verdict we draw from that run rather than from the machinery-validation sample, and it bounds the effect we could have seen at this sample size rather than excluding one. Second, the meaning-preserving guarantee rests on an automatic gate; a human equivalence audit is needed before it is fully established. Whether a simple reformulation-averaging defense confers measurable robustness is an open question we frame but do not yet settle. The experiment is specified and has not been run. We have specified it so that it can fail: the d -averaged arm must beat a cost-matched $d=1$ fresh-redraw control by a per-question paired margin whose 95% interval excludes zero, and an interval covering zero means the defense bought the attacker’s query cost rather than robustness. Taken together, the results argue that “semantic” uncertainty inherits the blind spots of whatever model defines its notion of meaning. They also argue that detectors built on a learned equivalence relation should be evaluated as the adversarial targets they now are.

Data and Code Availability

Code, analysis scripts, the dated reports of record that own the numbers above, and the plotted points behind every figure are in the repository, under the MIT licence:

<https://github.com/ScriptSampler/red-team-semantic-entropy>

What a clone buys differs sharply between the two halves of the headline, and we would rather say so than let a URL imply more than it delivers. The combinatorial half reproduces from a bare clone with no data, no model and no GPU: the 39 attainable values at $N=10$, the two of them in the top tenth of the range, the atom at $\ln 10$, and the corresponding counts at $N=20$ and $N=40$ are an enumeration over integer partitions, pinned on CPU in seconds against an independent implementation. The read-only checkers that verify the population label on every rate we print, and that re-derive the printed quantities with no producing artifact of their own, run in a fresh clone as well. The empirical half does not. Every measured rate here is computed from a sample cache of 2000 TriviaQA rc.nocontext questions with ten Llama-3.1-8B-Instruct generations each, 4.75 MB, which lives outside the repository tree and is not currently distributed; the null control’s per-target checkpoints are written into the tree and then excluded from it. A clone therefore carries every derived report and each figure’s plotted points as a CSV, so the arithmetic is auditable from the run outputs onward, but it cannot re-run the pipeline that produced them, and we claim no more reproducibility than that. ARTIFACT_AVAILABILITY.md in the repository names the files a clone is missing, with their sizes and SHA-256 digests. Model weights and datasets are obtained from their upstream sources under their own licences and are not redistributed here.

References

- [1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.
- [2] Phil Blandfort and Robert Graham. Red-teaming activation probes using prompted llms. *arXiv preprint arXiv:2511.00554*, 2025.
- [3] Yize Cheng, Vinu Sankar Sadasivan, Mehrdad Saberi, Shoumik Saha, and Soheil Feizi. Adversarial paraphrasing: A universal attack for humanizing ai-generated text. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [4] Alexandra Chouldechova, A. Feder Cooper, Solon Barocas, Abhinav Palia, Dan Vann, and Hanna Wallach. Comparison requires valid measurement: Rethinking attack success rate comparisons in AI red teaming. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [5] Hao Fang, Jiawei Kong, Tianqu Zhuang, Yixiang Qiu, Kuofeng Gao, Bin Chen, Shu-Tao Xia, Yaowei Wang, and Min Zhang. Your language model can secretly write like humans: Contrastive paraphrase attacks on LLM-generated text detectors. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8585–8602, 2025. doi: 10.18653/v1/2025.emnlp-main.433.
- [6] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630, 2024.
- [7] Tom Fawcett. An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874, 2006.
- [8] John Hughes, Sara Price, Aengus Lynch, Rylan Schaeffer, Fazl Barez, Sanmi Koyejo, Henry Sleight, Erik Jones, Ethan Perez, and Mrinank Sharma. Best-of-N jailbreaking. *arXiv preprint arXiv:2412.03556*, 2024.
- [9] Masahiro Kaneko. Paraphrasing adversarial attack on llm-as-a-reviewer. *arXiv preprint arXiv:2601.06884*, 2026.
- [10] Reza Khanmohammadi, Ivan Brugere, Simerjot Kaur, Charese H. Smiley, Kundan Thind, and Mohammad M. Ghassemi. Model confidence under answer-preserving attacks: An informativeness–manipulability frontier. *arXiv preprint arXiv:2608.06571*, 2026.

- [11] Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in llms. In *ICML Workshop on Foundation Models in the Wild*, 2024.
- [12] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *International Conference on Learning Representations (ICLR)*, 2023.
- [13] Erich L. Lehmann and Joseph P. Romano. *Testing Statistical Hypotheses*. Springer, New York, third edition, 2005.
- [14] Buyun Liang, Liangzu Peng, Jinqi Luo, Darshan Thaker, Kwan Ho Ryan Chan, and René Vidal. SECA: Semantically equivalent and coherent attacks for eliciting llm hallucinations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [15] Buyun Liang, Jinqi Luo, Liangzu Peng, Kwan Ho Ryan Chan, Darshan Thaker, Kaleab A. Kinfu, Fengrui Tian, Hamed Hassani, and René Vidal. REALISTA: Realistic latent adversarial attacks that elicit llm hallucinations. In *International Conference on Machine Learning (ICML)*, 2026.
- [16] Josué Martínez-Martínez, John T. Holodnak, Olivia Brown, Sheida Nabavi, Derek Aguiar, and Allan Wollaber. Improving vision model robustness against misclassification and uncertainty attacks via underconfidence adversarial training. In *Proceedings of the 7th Northern Lights Deep Learning Conference (NLDL)*, volume 307 of *PMLR*, pages 274–286, 2026.
- [17] Lucas H. McCabe, Rimón Melamed, Thomas Hartvigsen, and H. Howie Huang. Estimating semantic alphabet size for LLM uncertainty quantification. *arXiv preprint arXiv:2509.14478*, 2025.
- [18] Nay Myat Min, Long H. Pham, Hongyu Zhang, and Jun Sun. CORVUS: Red-teaming hallucination detectors via internal signal camouflage in large language models. *arXiv preprint arXiv:2601.14310*, 2026.
- [19] Jerzy Neyman and Egon S. Pearson. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society of London. Series A*, 231:289–337, 1933.
- [20] Alexander Nikitin, Jannik Kossen, Yarin Gal, and Pekka Marttinen. Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024.
- [21] Stephen Obadinma and Xiaodan Zhu. On the robustness of verbal confidence of LLMs in adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [22] Stephen Obadinma, Xiaodan Zhu, and Hongyu Guo. Calibration attacks: A comprehensive study of adversarial attacks on model confidence. *Transactions on Machine Learning Research*, 2024. *arXiv:2401.02718*.
- [23] Hongxing Pan, Yingying Guo, Wenqing Kuang, and Jiashi Lu. Mind the unseen mass: Unmasking llm hallucinations via soft-hybrid alphabet estimation. *arXiv preprint arXiv:2604.19162*, 2026.
- [24] Foster Provost and Tom Fawcett. Robust classification for imprecise environments. *Machine Learning*, 42(3):203–231, 2001.
- [25] Giulio Rossolini. Worst-case vs. random perturbations in robustness evaluation. *arXiv preprint arXiv:2601.14519*, 2026.

- [26] Jonathan Rusert. RedHerring attack: Testing the reliability of attack detection. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- [27] Ao Sun, Tian Sun, and Jiaxing Geng. The score granularity gap in black-box LLM classification: A comparative study of confidence constructions. *arXiv preprint arXiv:2606.22179*, 2026.
- [28] Chaodong Tong, Qi Zhang, Lei Jiang, Yanbing Liu, Nannan Sun, and Wei Li. Semantic reformulation entropy for robust hallucination detection in qa tasks. *arXiv preprint arXiv:2509.17445*, 2025.
- [29] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005.
- [30] Mingyi Wang, Zhuoer Shen, Yuheng Bu, and Shaofeng Zou. Detector-evasive llm paraphrasing via constrained policy optimization. *arXiv preprint arXiv:2606.00392*, 2026.
- [31] Qingcheng Zeng, Mingyu Jin, Qinkai Yu, Zhenting Wang, Wenyue Hua, Zihao Zhou, Guangyan Sun, Yanda Meng, Shiqing Ma, Qifan Wang, Felix Juefei-Xu, Kaize Ding, Fan Yang, Ruixiang Tang, and Yongfeng Zhang. Uncertainty is fragile: Manipulating uncertainty in large language models. *arXiv preprint arXiv:2407.11282*, 2024.
- [32] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.
- [33] Puzhuo Zheng and Hasan Kurban. It’s the decoding format, not the perturbation: Auditing consistency-based selection for vision-language test-time scaling. *arXiv preprint arXiv:2608.01207*, 2026.